

Metal–Oxide–Semiconductor Field-Effect Transistors (MOSFETs)

Myar Mohamed 202300928 – Fatima Magdy 202200813 – Sondos Ahmed 202300162 – Habeba Sherif 202300930 – Mohamed AbdelRahman 202301006 – Ahmed AbdelNasser 202000783

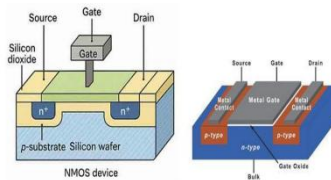
1. MOSFET Basic Structure

The Metal–Oxide–Semiconductor Field-Effect Transistor (MOSFET) is a four-terminal semiconductor device consisting of a Gate, Source, Drain, and Body (substrate). The device operates by modulating the conductivity of a semiconductor channel using an electric field applied through a thin gate-oxide layer. MOSFETs are foundational to modern CMOS technology due to their scalability, low power consumption, and compatibility with integrated circuit fabrication processes.

2. 1.1 Physical Layers and Device Geometry

A MOSFET is formed using a polysilicon or metal gate placed on top of a thin SiO₂ gate oxide, which in turn lies on a lightly doped silicon substrate.

- For an NMOS, the substrate is p-type, and the Source and Drain are heavily doped n⁺ regions.
- For a PMOS, the substrate is n-type, with p⁺ Source and Drain regions. This structure creates the essential metal–oxide–semiconductor (MOS) capacitor, which governs the electrostatics of channel formation[7], [11].



1.2 Source/Drain Formation and Junctions

The Source and Drain terminals are created by heavy doping to form low-resistance ohmic contacts. Their junctions with the substrate must be shallow to reduce parasitic resistance and to suppress short-channel effects at advanced nodes. Long-channel MOSFETs have nearly symmetric source and drain regions, while scaled technologies may use lightly doped drain (LDD) extensions to reduce electric-field peaks near the drain [7], [8].

1.3 Channel Region

The channel lies directly under the gate oxide and connects the source to the drain when an appropriate gate voltage is applied. The channel length (L) is the horizontal distance between source and drain, while the width (W) determines the current-carrying capability. The channel is normally non-conductive; conductivity appears when an inversion layer forms at the semiconductor surface [7], [9].

1.4 MOSFET as a MOS Capacitor

The MOSFET relies on the MOS capacitor effect. As the gate voltage varies, the surface of the substrate transitions through three electrostatic states:

1. Accumulation: Majority carriers gather at the surface.
2. Depletion: Majority carriers are repelled, leaving fixed ionized dopants.
3. Inversion: Minority carriers accumulate, forming a conductive channel under the gate.

This transition from depletion to inversion is essential for channel formation and is described mathematically in the MOS capacitor surface-potential relations derived in [10]

2. MOSFET Working Principle

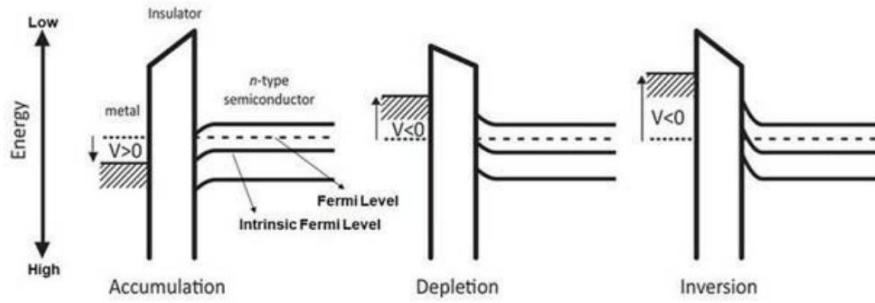
The MOSFET operates by controlling the flow of carriers from Source to Drain through an inversion channel created by the gate voltage. The gate voltage modulates the electrostatic potential at the semiconductor surface, thereby determining whether a conductive channel exists and how strongly it conducts.

2.1 Channel Formation and Inversion Mechanism

When a gate-to-source voltage (V_{GS}) is applied:

1. A positive V_{GS} in an NMOS attracts electrons to the Si–SiO₂ interface.
2. The substrate first enters depletion, widening the depletion region.
3. When the surface potential reaches 2 ϕ_F , an inversion layer of electrons forms.
4. This inversion layer becomes a conductive channel between Source and Drain.

This condition occurs when V_{GS} = V_{th}, where the threshold voltage depends on oxide thickness, substrate doping, flatband voltage, oxide charge, and substrate bias, as derived rigorously in [10].



2.2 Threshold Voltage (V_{th})

Threshold voltage is the gate voltage required to reach strong inversion. It is expressed by:

$$V_{th} = V_{FB} + 2\phi_F + \frac{\sqrt{2\epsilon_s q N_A (2\phi_F + V_{SB})}}{C_{ox}}$$

Each term corresponds to a physical component of the MOS capacitor:

- V_{FB} : Flatband voltage.
- $2\phi_F$: Required surface potential for inversion.
- Depletion-charge term: Depends on substrate doping.
- C_{ox} : Governs gate control effectiveness.
- Body-effect term (V_{SB}): Increases V_{th} when substrate is reverse-biased. These relationships are derived in detail in [10].

2.3 MOSFET Operation Regions

Once a channel forms, the MOSFET moves through three operating regions depending on V_{GS} and V_{DS} [9], [11]:

(a) Cutoff Region

- Condition: $V_{GS} < V_{th}$
- No inversion layer $\rightarrow I_D \approx 0$
- MOSFET is OFF. (b) Linear / Triode Region
- Conditions: $V_{GS} > V_{th}$ and small V_{DS}
- Channel is uniform across the length
- Current approximately linear in V_{DS}
- Described by the drift-current equation:

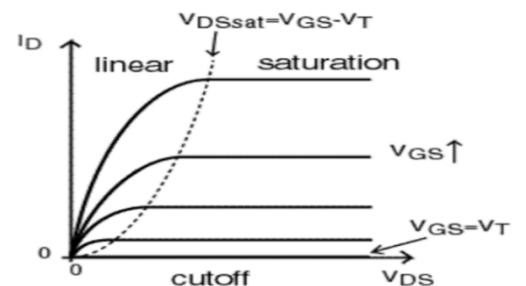
$$I_D = \mu C_{ox} \frac{W}{L} [(V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2}]$$

(c) Saturation Region

- Condition: $V_{DS} \geq V_{GS} - V_{th}$
- Channel near the drain becomes pinched off
- Increasing V_{DS} further does not significantly increase I_D
- Saturation current:

$$I_{Dsat} = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

These models correspond to long-channel behavior; short-channel devices introduce velocity saturation and other high-field effects discussed in [11].

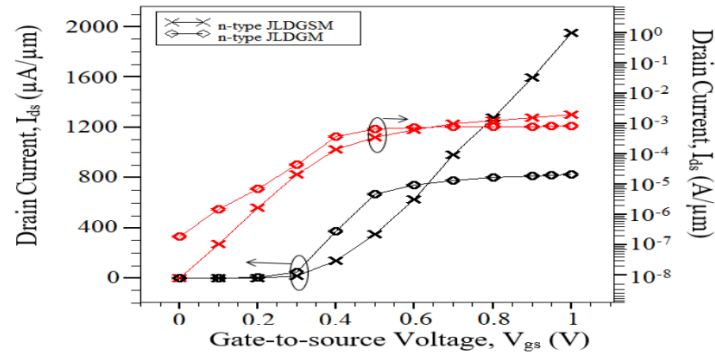


2.3 TRANSFER CHARACTERISTICS (ID–VGS BEHAVIOR)

AT A FIXED V_{DS} , THE RELATIONSHIP BETWEEN I_D AND V_{GS} SHOWS:

- SUBTHRESHOLD REGION: EXPONENTIAL INCREASE IN CURRENT.
- THRESHOLD TRANSITION: ONSET OF STRONG INVERSION.
- STRONG INVERSION: I_D RISES QUADRATICALLY OR LINEARLY DEPENDING ON V_{DS} .

THIS CURVE IS FUNDAMENTAL FOR THRESHOLD EXTRACTION AND DEVICE DESIGN [11].



3. KEY DESIGN PARAMETERS IN MOSFET

DESIGN, PARAMETERS ARE GENERALLY CATEGORIZED INTO GEOMETRIC/PHYSICAL PARAMETERS (INPUTS DEFINED IN THE FABRICATION PROCESS OR TCAD DECK) AND ELECTRICAL/PERFORMANCE PARAMETERS (OUTPUTS SIMULATE TO VERIFY THE DEVICE WORKS).

3.1. GEOMETRIC & PHYSICAL PARAMETERS (INPUTS)

THESE ARE THE VARIABLES THAT CAN PHYSICALLY CHANGE DURING THE DEVICE STRUCTURE DEFINITION IN SILVACO.

CHANNEL LENGTH (L):

DEFINITION: THE DISTANCE BETWEEN THE SOURCE AND DRAIN METALLURGICAL JUNCTIONS.

SIGNIFICANCE: THIS IS THE MOST CRITICAL SCALING PARAMETER. REDUCING L INCREASES CURRENT (I_{DS}) AND SWITCHING SPEED BUT INTRODUCES SHORT CHANNEL EFFECTS (SCE).

CHANNEL WIDTH (W):

DEFINITION: THE LATERAL DIMENSION OF THE TRANSISTOR PERPENDICULAR TO CURRENT FLOW.

SIGNIFICANCE: DIRECTLY PROPORTIONAL TO THE CURRENT CAPABILITY ($I_{DS} \propto W$).

INCREASING W INCREASES DRIVE CURRENT BUT ALSO INCREASES PARASITIC CAPACITANCE AND DEVICE AREA.

OXIDE THICKNESS (t_{ox}):

DEFINITION: THE THICKNESS OF THE SILICON DIOXIDE (SiO_2) INSULATING LAYER BETWEEN THE GATE AND THE CHANNEL.

SIGNIFICANCE: A THINNER OXIDE (t_{ox}) INCREASES THE GATE CAPACITANCE (C_{ox}), WHICH IMPROVES THE GATE'S CONTROL OVER THE CHANNEL AND SUPPRESSES LEAKAGE.

HOWEVER, IF TOO THIN, IT LEADS TO GATE TUNNELING LEAKAGE.

DOPING CONCENTRATION :

DEFINITION: THE CONCENTRATION OF IMPURITIES (ACCEPTORS/DONORS) IN THE SUBSTRATE, SOURCE, AND DRAIN.

SIGNIFICANCE: HIGHER SUBSTRATE DOPING HELPS CONTROL THE THRESHOLD VOLTAGE (V_{th}) AND PREVENTS PUNCH-THROUGH, BUT LOWERS CARRIER MOBILITY DUE TO IMPURITY SCATTERING.

2. ELECTRICAL PERFORMANCE PARAMETERS (OUTPUTS)

THESE ARE THE METRICS EXTRACTED FROM SIMULATION CURVES ($I_D - V_g$ and $I_D - V_d$).

THRESHOLD VOLTAGE : THE MINIMUM GATE VOLTAGE REQUIRED TO CREATE A STRONG INVERSION LAYER (CHANNEL) CONNECTING SOURCE AND DRAIN.

SUBTHRESHOLD SWING (SS): A MEASURE OF HOW FAST THE TRANSISTOR TURNS OFF.

IDEALLY, IT IS 60 mV/DECADE AT ROOM TEMPERATURE. HIGH SS INDICATES HIGH LEAKAGE POWER.

ON-CURRENT (I_{on}) VS. OFF-CURRENT (I_{off}):

I_{on} : CURRENT WHEN THE DEVICE IS FULLY ON (HIGH PERFORMANCE).

I_{off} : LEAKAGE CURRENT WHEN THE DEVICE IS OFF (POWER CONSUMPTION).

A HIGH I_{on}/I_{off} RATIO IS THE ULTIMATE DESIGN GOAL.

TRANSCONDUCTANCE (g_m): REPRESENTS THE GAIN OF THE DEVICE.

IT MEASURES HOW MUCH THE DRAIN CURRENT CHANGES FOR A SMALL CHANGE IN GATE VOLTAGE ($\partial I_d/\partial V_g$).

4. MOSFETS APPLICATIONS:

4.1. SWITCHING: A MAJOR USE OF MOSFETs IN POWER ELECTRONICS IS THEIR ROLE AS HIGH-SPEED SWITCHES.

THEIR ABILITY TO TURN ON AND OFF RAPIDLY ENABLES OPERATION AT HIGH FREQUENCIES WHILE MINIMIZING POWER LOSSES. IN ADDITION, MOSFETs CAN WITHSTAND LARGE CURRENTS AND VOLTAGES, MAKING THEM WELL-SUITED FOR DEMANDING, HIGH-POWER APPLICATIONS. **DC-DC CONVERTERS** MODIFY A DC VOLTAGE FROM ONE LEVEL TO ANOTHER. A MOSFET OPERATES AS A SWITCHING ELEMENT THAT ALTERNATELY ALLOWS AND BLOCKS CURRENT, CONTROLLING HOW AN INDUCTOR OR CAPACITOR STORES AND RELEASES ENERGY EACH CYCLE. THE FINAL OUTPUT VOLTAGE IS SET BY THE DUTY CYCLE OF THE SWITCH, WHICH IS THE PROPORTION OF TIME IT REMAINS ON VERSUS OFF [5]. **MOTOR CONTROL** CIRCUITS MANAGE BOTH THE SPEED AND DIRECTION OF ELECTRIC MOTORS. A MOSFET FUNCTIONS AS A SWITCHING DEVICE THAT REGULATES THE CURRENT PASSING THROUGH THE MOTOR'S WINDINGS, GENERATING THE MAGNETIC FIELDS THAT DRIVE THE SHAFT'S ROTATION. BY ADJUSTING THE SWITCHING FREQUENCY AND CHANGING THE POLARITY, THE MOTOR'S SPEED AND ROTATIONAL DIRECTION CAN BE PRECISELY CONTROLLED [5].

INVERTERS ARE CIRCUITS THAT TRANSFORM DC VOLTAGE INTO AC VOLTAGE. A MOSFET ACTS AS A HIGH-SPEED SWITCH THAT RAPIDLY REVERSES THE POLARITY OF THE DC SUPPLY, PRODUCING AN AC WAVEFORM AT THE OUTPUT. BY MODIFYING THE SWITCHING SEQUENCE, BOTH THE FREQUENCY AND AMPLITUDE OF THE RESULTING AC SIGNAL CAN BE CONTROLLED.

4.2 AMPLIFICATION:

MOSFETs ARE ALSO WIDELY USED AS AMPLIFIERS IN ANALOG CIRCUITS. THEY CAN BOOST A SMALL INPUT SIGNAL BY USING A SMALL VOLTAGE CHANGE TO CONTROL A MUCH LARGER OUTPUT CURRENT. DEPENDING ON THE APPLIED INPUT AND OUTPUT VOLTAGES, A MOSFET CAN FUNCTION IN VARIOUS OPERATING REGIONS—SUCH AS CUT-OFF, LINEAR, OR SATURATION—TO ACHIEVE DIFFERENT TYPES OF AMPLIFICATION BEHAVIOR.

RADIO-FREQUENCY AMPLIFIERS STRENGTHEN SIGNALS IN THE RF RANGE, COVERING RADIO WAVES AND MICROWAVES. MOSFETs ARE USED TO INCREASE THE POWER OF THESE SIGNALS, IMPROVING BOTH TRANSMISSION AND RECEPTION. THEIR HIGH SWITCHING SPEED MAKES THEM WELL-SUITED FOR OPERATION AT THESE ELEVATED FREQUENCIES[5].

AUDIO AMPLIFIERS BOOST SIGNALS WITHIN THE AUDIO-FREQUENCY RANGE, SUCH AS SPEECH OR MUSIC. A MOSFET CAN SERVE AS THE AMPLIFYING DEVICE, RAISING THE SIGNAL'S STRENGTH TO IMPROVE SOUND QUALITY OR LOUDNESS FOR PLAYBACK OR RECORDING. ITS HIGH INPUT IMPEDANCE AND LOW OUTPUT IMPEDANCE HELP ACHIEVE LOW DISTORTION AND MINIMAL NOISE.

SENSOR AMPLIFIERS STRENGTHEN THE SMALL SIGNALS PRODUCED BY DEVICES LIKE TEMPERATURE, PRESSURE, LIGHT, OR MOTION SENSORS. USING A MOSFET AS THE AMPLIFIER INCREASES THE SENSOR'S SENSITIVITY AND ACCURACY, ENHANCING MEASUREMENT PERFORMANCE. THANKS TO ITS SIMPLE AND DURABLE DESIGN, A MOSFET CAN PROVIDE AMPLIFICATION WITH LOW POWER USAGE AND HIGH RELIABILITY.

5. MOSFET TYPES / VARIATIONS ENHANCEMENT-TYPE MOSFET

IN AN ENHANCEMENT MOSFET, THE CHANNEL DOES NOT EXIST AT ZERO GATE BIAS, SO NO CURRENT CAN INITIALLY FLOW BETWEEN THE SOURCE AND DRAIN. WHEN A POSITIVE GATE VOLTAGE EXCEEDING THE THRESHOLD VOLTAGE (THE GATE VOLTAGE THAT TURNS THE MOSFET ON) IS APPLIED, AN INDUCED CHANNEL FORMS DUE TO THE ELECTRIC FIELD FROM THE GATE. THIS "ENHANCES" THE CONDUCTIVITY OF THE DEVICE AND ALLOWS CURRENT TO FLOW. ENHANCEMENT MOSFETs ARE THE MOST COMMONLY USED TYPE IN MODERN DIGITAL CIRCUITS.

- DEPLETION-TYPE MOSFET

A DEPLETION MOSFET HAS A CHANNEL ALREADY FORMED AT ZERO GATE VOLTAGE, MEANING IT CAN CONDUCT CURRENT EVEN WITHOUT ANY APPLIED GATE BIAS. APPLYING A NEGATIVE GATE VOLTAGE DECREASES THE CHANNEL'S CARRIER CONCENTRATION, EFFECTIVELY NARROWING THE CHANNEL AND REDUCING THE CURRENT FLOW. THESE DEVICES ARE LESS COMMONLY USED IN TYPICAL INTEGRATED CIRCUITS.

5.1 MOSFET VARIATIONS: ENHANCEMENT AND DEPLETION TYPES (N-CHANNEL & PCHANNEL)

MOSFETs COME IN TWO MAIN OPERATING MODES: 1-ENHANCEMENT AND 2-DEPLETION, AND EACH MODE CAN EXIST IN N-CHANNEL OR P-CHANNEL FORM. THEIR BEHAVIOR DEPENDS ON WHETHER A CHANNEL EXISTS NATURALLY OR MUST BE CREATED USING THE GATE VOLTAGE.

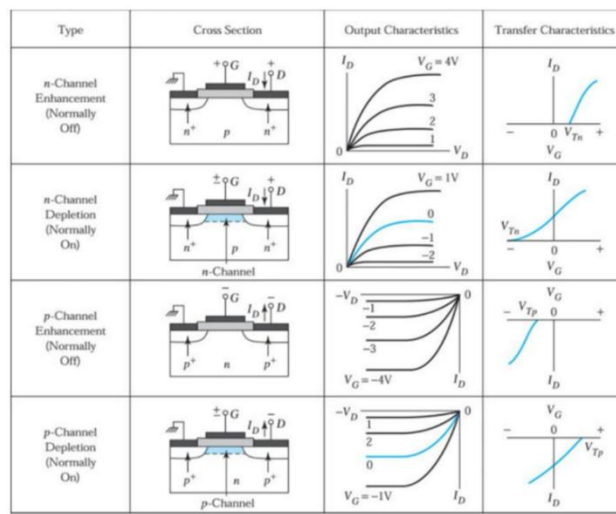


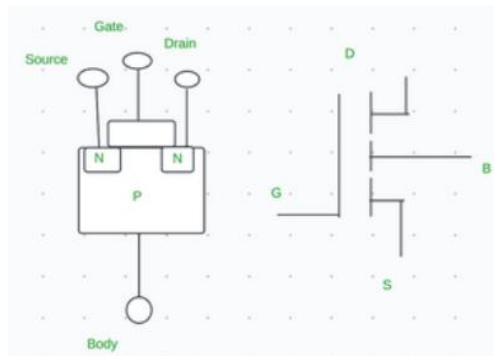
Fig. 26 Cross section, output, and transfer characteristics of four types of MOSFETs.

Source: S. M. Sze and M. K. Lee, Semiconductor Devices: Physics and Technology, 3rd Edition, Fig. 26.

5.1.1. N-CHANNEL ENHANCEMENT MOSFET (NORMALLY OFF)

OPERATION 12

- THE DEVICE INITIALLY HAS NO CHANNEL, SO NO DRAIN CURRENT FLOWS AT $V_G=0V$.
- APPLYING A POSITIVE GATE VOLTAGE ATTRACTS ELECTRONS TO THE SEMICONDUCTOR SURFACE.
- ONCE THE GATE VOLTAGE EXCEEDS THE THRESHOLD VOLTAGE V_{TN} , AN INVERSION LAYER OF ELECTRONS FORMS, CREATING AN N-CHANNEL.
- INCREASING V_G WIDENS THE CHANNEL, ALLOWING MORE CURRENT.



APPLICATIONS

- DOMINANT DEVICE IN DIGITAL ICs AND POWER ELECTRONICS.
 - HIGH EFFICIENCY DUE TO FAST ELECTRON MOBILITY.
- #### 5.1.2. N-CHANNEL DEPLETION MOSFET (NORMALLY ON) OPERATION

- A CONDUCTIVE N-CHANNEL IS ALREADY PRESENT AT $V_G=0V$. THE DEVICE NATURALLY CONDUCTS.
- TO REDUCE THE CURRENT, A NEGATIVE GATE VOLTAGE REMOVES ELECTRONS (DEPLETION), NARROWING THE CHANNEL.
- A POSITIVE GATE VOLTAGE ENHANCES CONDUCTION BY INCREASING ELECTRON DENSITY.

APPLICATIONS

- FUNCTIONED HISTORICALLY IN ANALOG CIRCUITS, LOADS, AND CURRENT SOURCES.
 - LESS COMMON IN MODERN CMOS.
- #### 5.1.3. P-CHANNEL ENHANCEMENT MOSFET (NORMALLY OFF) OPERATION
- BUILT ON AN N-TYPE SUBSTRATE WITH NO CONDUCTING CHANNEL AT ZERO GATE BIAS.
 - APPLYING A NEGATIVE GATE VOLTAGE ATTRACTS HOLES AND FORMS A P-CHANNEL ONCE THE GATE VOLTAGE CROSSES V_{TP} (THRESHOLD).
 - INCREASING THE MAGNITUDE OF NEGATIVE V_G INCREASES HOLE CONCENTRATION AND CURRENT.

APPLICATIONS

- COMMON IN HIGH-SIDE SWITCHING, BATTERY-POWERED DEVICES, AND POWER MANAGEMENT CIRCUITS.

5.1.4. P-CHANNEL DEPLETION MOSFET (NORMALLY ON) OPERATION

- A P-TYPE CHANNEL EXISTS NATURALLY AT ZERO GATE VOLTAGE.
- CURRENT FLOWS USING HOLES AS MAJORITY CARRIERS.
- APPLYING A POSITIVE GATE VOLTAGE REPELS HOLES, REDUCING CHANNEL WIDTH AND LOWERING CURRENT.
- APPLYING A NEGATIVE GATE VOLTAGE ENHANCES CONDUCTION.

APPLICATIONS

- RARE TODAY; SOMETIMES USED IN ANALOG CIRCUITS AS VOLTAGE-CONTROLLED RESISTORS.

5.5.5. N-CHANNEL VS. P-CHANNEL MOSFETS N-CHANNEL MOSFETS

- MAJORITY CARRIERS: ELECTRONS (HIGH MOBILITY).
- PROVIDE LOWER ON-RESISTANCE, FASTER SWITCHING, AND BETTER EFFICIENCY.
- IDEAL FOR HIGH-CURRENT, HIGH-FREQUENCY, AND POWER-SWITCHING APPLICATIONS.

P-CHANNEL MOSFETS

- MAJORITY CARRIERS: HOLES (LOWER MOBILITY).
- EXHIBIT HIGHER RESISTANCE AND SLOWER SWITCHING
- USED PRIMARILY FOR HIGH-SIDE SWITCHING, REVERSE POLARITY PROTECTION, AND BATTERY SWITCHING.

6. CHALLENGES IN MODERN SHORT-CHANNEL DEVICES

WHEN THE CHANNEL LENGTH (L) BECOMES COMPARABLE TO THE DEPLETION LAYER WIDTHS OF THE SOURCE AND DRAIN, THE "LONG CHANNEL" APPROXIMATIONS FAIL, AND SHORT CHANNEL EFFECTS (SCEs) APPEAR.

DRAIN-INDUCED BARRIER LOWERING (DIBL)

THE ISSUE: IN A LONG CHANNEL, ONLY THE GATE CONTROLS THE CHANNEL BARRIER. IN A SHORT CHANNEL, THE DRAIN VOLTAGE IS CLOSE ENOUGH TO THE SOURCE THAT THE DRAIN'S ELECTRIC FIELD PENETRATES THE CHANNEL AND LOWERS THE POTENTIAL BARRIER AT THE SOURCE.

THE RESULT: THE THRESHOLD VOLTAGE (V_{th}) DECREASES AS DRAIN VOLTAGE (V_d) INCREASES. THE GATE LOSES EXCLUSIVITY IN CONTROLLING THE CHANNEL, LEADING TO HIGHER LEAKAGE.

15.2. VELOCITY SATURATION

THE ISSUE: IN CLASSIC PHYSICS, CARRIER VELOCITY IS PROPORTIONAL TO THE ELECTRIC FIELD ($v = \mu * E$). IN SHORT DEVICES, THE ELECTRIC FIELD IS MASSIVE (VOLTAGE/LENGTH).

THE RESULT: CARRIERS REACH A "SPEED LIMIT" (SCATTERING LIMITS THEIR MAXIMUM SPEED). CURRENT NO LONGER INCREASES QUADRATICALLY WITH VOLTAGE; IT BECOMES LINEAR, LIMITING THE SPEED BOOST EXPECTED FROM SCALING.

3. HOT CARRIER INJECTION (HCI) THE ISSUE: ELECTRONS GAINING HIGH KINETIC ENERGY FROM THE HIGH ELECTRIC FIELD NEAR THE DRAIN ARE CALLED "HOT CARRIERS."

THE RESULT: THESE HIGH-ENERGY ELECTRONS CAN CRASH INTO THE GATE OXIDE INTERFACE, GETTING TRAPPED OR BREAKING CHEMICAL BONDS. OVER TIME, THIS SHIFTS THE V_{th} AND DEGRADES THE DEVICE RELIABILITY (AGING). 4. CHANNEL LENGTH MODULATION (CLM)

THE ISSUE: AS DRAIN VOLTAGE INCREASES, THE DEPLETION REGION AT THE DRAIN EXPANDS, SHORTENING THE EFFECTIVE CHANNEL LENGTH.

THE RESULT: CURRENT DOES NOT SATURATE PERFECTLY FLAT; IT KEEPS CREEPING UP AS V_d INCREASES, LOWERING THE OUTPUT RESISTANCE OF THE TRANSISTOR.

QUANTUM TUNNELING (GATE LEAKAGE)

THE ISSUE: TO MAINTAIN CONTROL OVER SHORT CHANNELS, DESIGNERS REDUCE OXIDE THICKNESS (tox). WHEN $tox < 2$ NM, QUANTUM MECHANICS TAKES OVER.

THE RESULT: ELECTRONS CAN "TUNNEL" THROUGH THE INSULATION FROM THE GATE TO THE CHANNEL, CAUSING SIGNIFICANT POWER LOSS EVEN WHEN THE DEVICE IS STATIC.

6. REFERENCES:

- [1] C. HU, MODERN SEMICONDUCTOR DEVICES FOR INTEGRATED CIRCUITS. UPPER SADDLE RIVER, NJ, USA: PEARSON, 2010.
- [2] D. A. NEAMEN, SEMICONDUCTOR PHYSICS AND DEVICES: BASIC PRINCIPLES, 4TH ED. NEW YORK, NY, USA: MCGRAW-HILL, 2012.
- [3] R. H. DENNARD, F. H. GAENSSLEN, H. N. YU, V. L. RIDEOUT, E. BASSOUS, AND A. R. LEBLANC, "DESIGN OF ION-IMPLANTED MOSFETS WITH VERY SMALL PHYSICAL DIMENSIONS," IEEE JOURNAL OF SOLID-STATE CIRCUITS, VOL. 9, NO. 5, PP. 256–268, OCT. 1974.
- [4] D. J. FRANK, R. H. DENNARD, E. NOWAK, P. M. SOLOMON, Y. TAUR, AND H.-S. P. WONG, "DEVICE SCALING LIMITS OF SI MOSFETS AND THEIR APPLICATION DEPENDENCIES," PROCEEDINGS OF THE IEEE, VOL. 89, NO. 3, PP. 259–288, MAR. 2001.
- [5] ELECTRICAL4U. (2023). APPLICATIONS OF MOSFET: HOW THEY WORK AND WHERE THEY ARE USED.

[HTTPS://WWW.ELECTRICAL4U.COM/APPLICATIONS-OF-MOSFET/](https://www.electrical4u.com/applications-of-mosfet/) ELECTRICAL4U.COM

[6] S. M. SZE AND M. K. LEE, SEMICONDUCTOR DEVICES: PHYSICS AND TECHNOLOGY, 3RD ED. HOBOKEN, NJ, USA: WILEY, 2012. 16

[7] "BASIC MOSFET STRUCTURE." BHARATH UNIVERSITY.

[HTTPS://WWW.BHARATHUNIV.AC.IN/COLLEGES1/DOWNLOADS/COURSEWARE_ECE/NOTES/BEC010%20%20VLSI%20-%20NOTES.PDF](https://www.bharathuniv.ac.in/colleges1/downloads/courseware_ece/notes/BEC010%20%20VLSI%20-%20NOTES.PDF) [8] P. KOGGE, "CMOS VLSI DESIGN MOSFETs—A SLIDE 2," 2015. AVAILABLE:

[HTTPS://WWW3.ND.EDU/~KOGGE/COURSES/CSE40462-VLSI-FA18/WWW/PUBLIC/LECTURES/MOSFETs-A.PDF](https://www3.nd.edu/~kogge/courses/cse40462-VLSI-FA18/www/public/lectures/MOSFETs-A.PDF)

[10] GEEKSFORGEEKS, "WHAT IS MOSFET?," JAN. 28, 2024.

[HTTPS://WWW.GEEKSFORGEEKS.ORG/ELECTRONICSENGINEERING/MOSFET/](https://www.geeksforgeeks.org/electronicengineering/mosfet/)

[11] N. GOLDSMAN, "DERIVATION OF MOSFET THRESHOLD VOLTAGE FROM THE MOS CAPACITOR ENEE 313 NOTES PROF. NEIL GOLDSMAN." AVAILABLE: [HTTPS://USER.ENG.UMD.EDU/~NEIL/ENEE313/MOSFET_THRESHOLD.PDF](https://user.eng.umd.edu/~neil/ENEE313/MOSFET_THRESHOLD.PDF)

[12] SHANGHAI JIAO TONG UNIVERSITY, "LECTURE 14 MOSFET I-V CHARACTERISTICS," NOV. 2017. AVAILABLE:

[HTTPS://HSIC.SJTU.EDU.CN/ASSETS/USERFILES/SYS_EB538C1C-65FF-4E82-8E6AA1EF01127FED/FILES/LEC14 MOSFET I-V CHARACTERISTICS.PDF](https://hsic.sjtu.edu.cn/assets/userfiles/sys_eb538c1c-65ff-4e82-8e6aa1ef01127fed/files/lec14_mosfet_i-v_characteristics.pdf)

[13] "LECTURE 8 MOSFET(I) MOSFET I-V CHARACTERISTICS OUTLINE," 2007. AVAILABLE:

[HTTPS://WEB.MIT.EDU/6.012/WWW/SP0](https://web.mit.edu/6.012/www/SP0)

.

Introduction:

In this work, a p-channel MOSFET representative of a 90 nm CMOS technology node is modeled using the Silvaco ATLAS device simulator. The device operates at a supply voltage of $V_{DD} = 1.2$ V and has a gate length of 90 nm with a gate width of $10L$, following standard PMOS sizing conventions. A thin 2 nm gate oxide and a polysilicon gate are used to ensure strong electrostatic control of the channel.

The substrate is lightly doped n-type at $1 \times 10^{18} \text{ cm}^{-3}$, while the source and drain regions are heavily p-doped at $1 \times 10^{19} \text{ cm}^{-3}$, with a junction depth of 60 nm. A fixed oxide charge density of $Q_{ss} = +5 \times 10^{10} \text{ cm}^{-2}$ is included to account for interface effects, and the substrate depth is set to 200 nm. These parameters are kept constant to ensure that observed device behavior reflects intrinsic electrical and scaling effects rather than variations in material properties.

Code:

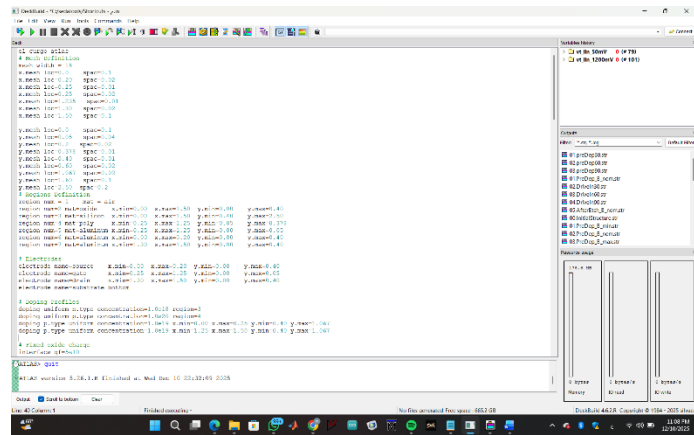


Fig.1

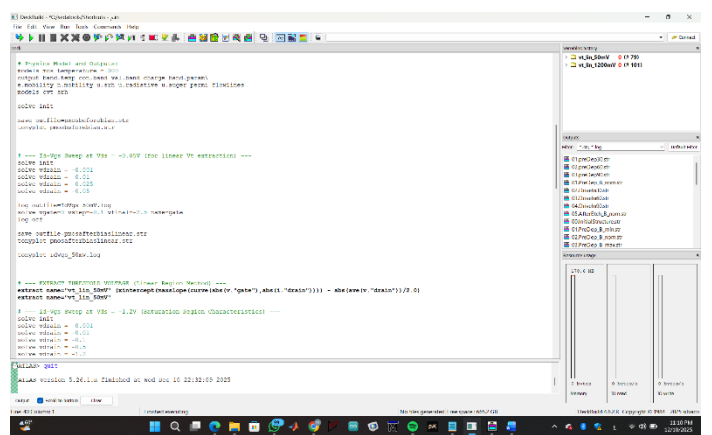


Fig.2

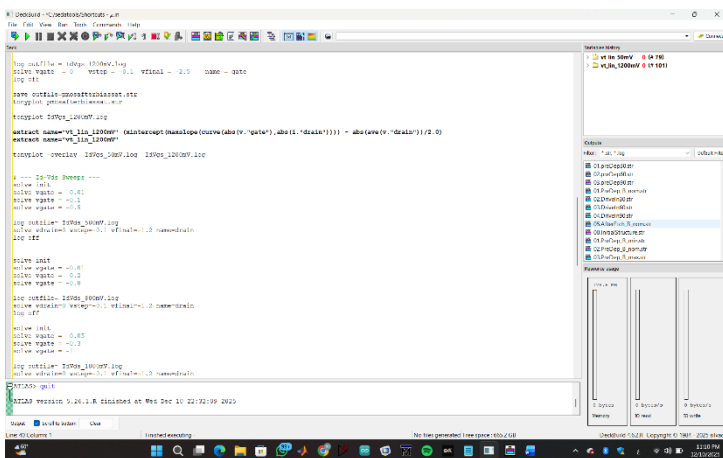


Fig.3

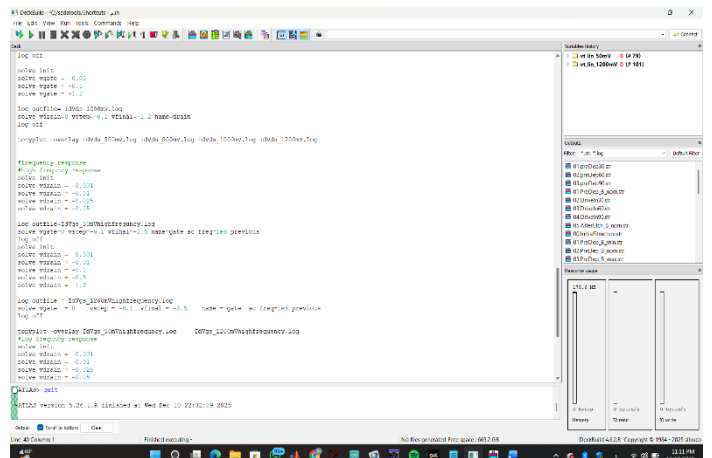


Fig.4

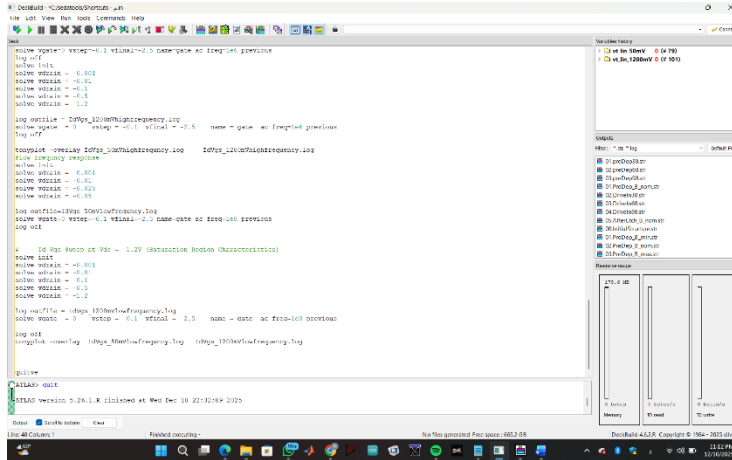


Fig.5

Code Explanation:

1. Mesh Definition

The simulation begins with defining a non-uniform mesh in both the x and y directions.

A finer mesh is used around critical regions such as the channel, oxide interface, and the source/drain junctions to ensure numerical accuracy, while a coarser mesh is used elsewhere to reduce computation time.

- $x.mesh$: Provides finer spacing (0.001 – $0.002 \mu\text{m}$) around the channel area ($-0.03 \mu\text{m}$ to $0.1 \mu\text{m}$).
- $y.mesh$: Uses tight spacing ($0.001 \mu\text{m}$) around the gate oxide and upper silicon region where strong electric fields exist.

2. Region Definitions

Each physical part of the MOSFET is defined as a separate region:

- Region 1 (oxide): The gate oxide layer.
- Region 2 (silicon): The silicon substrate where the channel is formed.
- Region 3 (poly gate): The polysilicon gate electrode.
- Region 4, 5, 6 (aluminum): Gate, drain, and source metal contacts.

Defining regions separately allows the simulator to assign different materials, doping profiles, and electrodes.

3. Electrode Specification

Electrodes are added to the physical boundaries of the gate, source, drain, and substrate:

- gate electrode sits on the polysilicon region.
- source and drain electrodes occupy the highly doped S/D regions.
- substrate contact is applied at the bottom of the silicon wafer.

These electrodes serve as terminals for applying voltages during the simulation (e.g., V_{gs} , V_{ds}).

4. Doping Profiles

The doping block specifies the electrical characteristics of the MOSFET:

- Channel doping:
 - Slightly lower n-type doping ($9 \times 10^{17} \text{ cm}^{-3}$) improves hole mobility.
- Poly gate doping:
 - n-type polysilicon doping ($7 \times 10^{18} \text{ cm}^{-3}$) sets appropriate gate work function.
- Source/Drain heavy p-type doping:
 - $5 \times 10^{19} \text{ cm}^{-3}$ to ensure low resistance and strong PMOS operation.

The abrupt transition between channel and S/D doping helps maintain high drive current.

5. Interface Traps and Gate Work Function

- interface $qf=5 \times 10^{10}$ introduces fixed oxide charges.
These are necessary to mimic realistic interface imperfections.
- workfunction=5.15 eV for the poly-gate sets the threshold voltage of the PMOS transistor.

6. Physical Models

The simulator activates several physical models:

- cvt – concentration-dependent mobility
- srh – Shockley–Read–Hall recombination
- fldmob – field-dependent mobility
- band.temp – temperature-dependent bandgap
- u.srh, u.auger, permi – additional recombination and mobility effects

These models ensure accurate current behavior under high electric fields and high temperature (500 K in this simulation).

7. Initial Solution

method newton autnrcarriers=2

solve init

ATLAS uses Newton's numerical method to obtain a stable initial operating point before applying external biases.

The structure is saved as:

save outfile=Optimized_Pmos_130nm.str

8. Id-Vgs Simulation ($V_{ds} = -1.2$ V)

A DC sweep is performed to extract the transfer characteristics:

1. Set drain voltage to -1.2 V (typical PMOS saturation condition).
2. Sweep the gate voltage from 0 to -2.5 V, in steps of -0.1 V.
3. Record the drain current during the sweep.

Output is saved in:

Optimized_IdVgs_1p2V.log

This file contains the I–V curve data used later for plotting and analysis.

9. Extracting ION and IOFF

ATLAS automatically reads the drain current from the log file and identifies:

- $I_{ON} \rightarrow$ maximum $|I_d|$ during strong inversion
- $I_{OFF} \rightarrow$ minimum $|I_d|$ in the subthreshold region

These parameters are essential to evaluate device performance, leakage, and overall technology scaling behavior.

10. Plotting the Results

tonyplot Optimized_IdVgs_1p2V.log

This command opens the I–V characteristics for visualization in TonyPlot.

Results:

PMOS IDS–VGS Characteristics and Operating Regions:

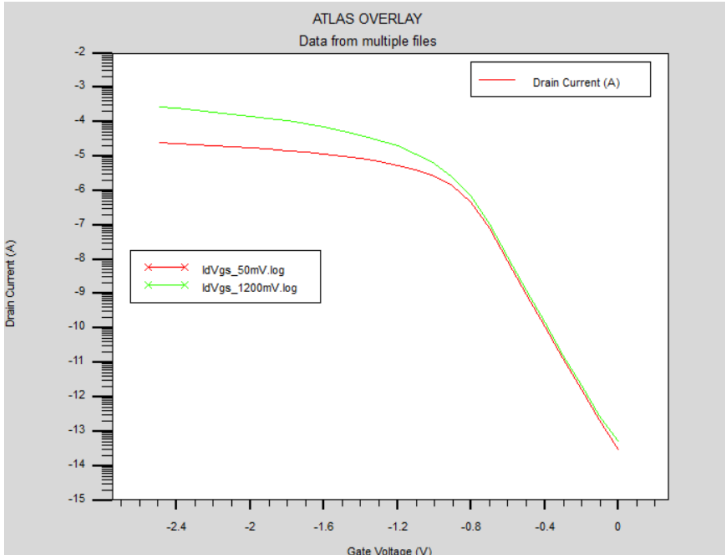


Fig.6 Changing of V_{SG} vs I_d at two values of V_{SD} 5mV and 1200 mV

The IDS–VGS plot illustrates the behavior of a pMOS transistor at two drain-to-source voltages: -0.05 V (red) and -1.2 V (green). As the gate voltage becomes more negative, the drain current increases accordingly. Using the linear extrapolation technique, the threshold voltage is approximated to be about -0.80 V.

- **Subthreshold Region:**

When $V_{SG} < |V_{TP}|$, the transistor remains off and carries only a very small leakage current. In this regime, I_D rises exponentially as the gate voltage becomes more negative. This behavior appears on the right-hand side of the graph, where both curves nearly overlap and the current is close to zero.

- **Linear (Triode) Region:**

Once $V_{SG} > |V_{TP}|$ and the drain-to-source voltage is small, the device enters the linear region. This is represented by the red curve at $V_{DS} = -0.05$ V. The current follows the relation:

$$I_{DS} = \mu_p C_{ox} \frac{W}{L} [(V_{SG} - |V_{TP}|) V_{SD} - \frac{V_{SD}^2}{2}]$$

- **Saturation Region:**

When the condition $V_{SD} > V_{SG} - |V_{TP}|$ is satisfied, the transistor operates in saturation. The green curve at $V_{DS} = -1.2$ V represents this mode, where the current levels off and becomes mostly independent of V_{DS} :

$$I_{DS} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{SG} - |V_{TP}|)^2$$

In summary, the plot clearly exhibits all expected operating regions of a pMOS transistor—subthreshold, linear, and saturation—showing how the device transitions between them as V_{GS} is swept.

And the threshold voltage extracted at both drain voltages, $V_{DS} = -0.05V$ and $V_{DS} = -1.2V$, is the same—approximately $-0.814V$. This aligns with the expected behavior of a long-channel pMOS transistor, where the threshold voltage depends mainly on the gate–source voltage (V_{SG}) and shows little to no dependence on the drain voltage (V_{DS}). The identical values obtained for different drain biases validate the extraction technique and further indicate that short-channel effects are not present in this device.

vt_lin_50mV	0
vt_lin_50mV	0.814363799162444
vt_lin_1200mV	0
vt_lin_1200mV	0.814363799162444

Fig. 7 Fig. 13. Threshold values for $V_{dS}=50mV$ and $V_{dS}=1200mV$

IDS–VDS Characteristics at Various Gate Biases

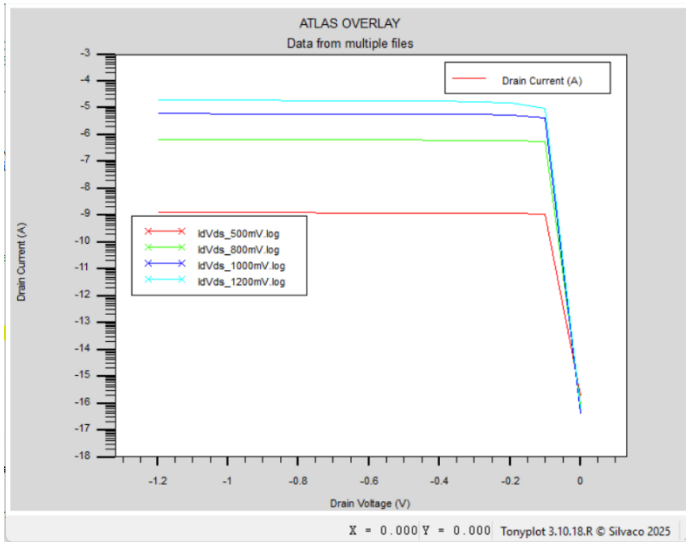


Fig. 8 V_{dS} vs I_d at values of $V_{gs}=500mV, 800mV, 1000mV, 1200mV$

The graph presents the relationship between the drain current (I_{DS}) and the drain voltage (V_{DS}) for a pMOS transistor under four gate-to-source voltages: $-0.5V$, $-0.8V$, $-1.0V$, and $-1.2V$.

For gate voltages that are less negative than the threshold voltage ($V_{TP} \approx -0.814V$), such as $V_{GS} = -0.5V$ and $-0.8V$, the transistor operates in the subthreshold region. As a result, the current remains nearly zero throughout the entire V_{DS} sweep, indicating that the device is off.

When the gate voltage becomes more negative than the threshold ($V_{GS} = -1.0V$ and $-1.2V$), the transistor switches on. At small values of V_{DS} (approximately 0 to $-0.3V$), the device behaves in the linear region, where I_{DS} increases roughly proportionally to V_{DS} .

As V_{DS} becomes more negative, the transistor transitions from the linear region into saturation, which starts when $V_{DS} < V_{SG} - |V_{TP}|$. For $V_{GS} = -1.0V$, this point occurs near $V_{DS} \approx -0.2V$, while for $V_{GS} = -1.2V$ it appears around $V_{DS} \approx -0.4V$. In saturation, changes in V_{DS} have little influence on I_{DS} , causing the current to level off.

Overall, the characteristics follow the expected long-channel pMOS behavior, clearly demonstrating subthreshold, linear, and saturation operation across different gate voltages.

Hole Current Density in the X and Y Directions

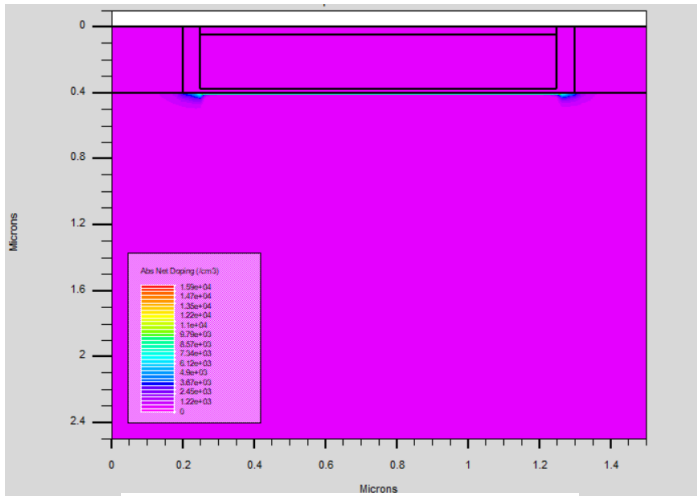


Fig.9 Electron Concentration

This figure shows the two-dimensional distribution of hole current density (A/cm^2) in the pMOS transistor under bias. The current flow pattern is influenced by the doping profile, the applied gate-source voltage, and the resulting electric fields in both the lateral (x) and vertical (y) directions.

- **X-axis:**

The hole current moves from the source to the drain just beneath the gate oxide. The negative gate bias creates a strong inversion layer, confining most of the current to the surface region. The highest current density appears under the gate, where the channel is formed and the electric field is strongest.

- **Y-axis:**

The gate generates a vertical electric field that pulls holes toward the silicon-oxide interface, restricting vertical current flow and keeping the conduction path close to the surface. Some vertical spreading occurs near the source and drain junctions due to diffusion.

Overall, the current distribution confirms accurate two-dimensional behavior, with dominant lateral conduction near the surface and appropriate current spreading around high-field junction areas.

Electric Field Distribution in the PMOS Structure

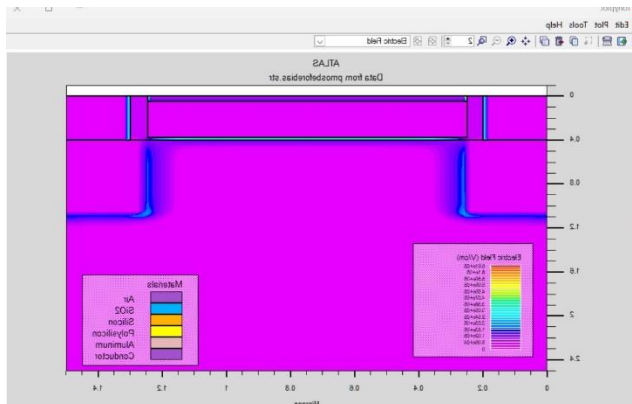


Fig.10 Electric field

The contour plot illustrates the electric field intensity within the PMOS device before applying any external bias. The electric field profile is dominated by both the **vertical (transverse)** and **lateral (longitudinal)** components, each playing a distinct role in determining the electrostatic behavior of the structure.

- **Transverse Electric Field (Y-Direction):**

The strongest electric field appears vertically beneath the gate oxide, where the gate stack meets the silicon substrate. This region shows a concentration of high-field colors (blue transitioning to cyan), indicating the abrupt permittivity change between SiO_2 and Si. Even without applied bias, this boundary generates localized field peaks due to material discontinuity and built-in potentials in the doped junctions.

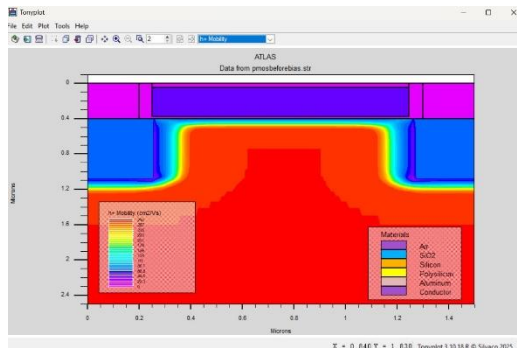
- **Longitudinal Electric Field (X-Direction):**

Along the horizontal channel direction, the electric field remains relatively uniform in the bulk substrate but becomes more pronounced near the **source/drain junctions**, where doping gradients create built-in electric fields. These regions appear with stronger field contours at the corners of the p^+ junctions, reflecting steep transitions in potential.

Overall, the electric field distribution confirms correct structural formation of the PMOS device. The field peaks near the oxide interface and junction corners are expected and indicate proper definition of the active channel region. This field configuration serves as the baseline for later biasing conditions, where stronger vertical and lateral fields will develop as the device transitions into linear and saturation operation.

Hole Mobility Distribution Before Applying Bias

The contour plot illustrates the hole mobility distribution in the PMOS structure before any external bias is applied. Since no gate or drain voltage is present, the electric field across the device is minimal, and the mobility is governed primarily by **doping concentration**, **material interfaces**, and **intrinsic lattice scattering** rather than field-dependent effects.



- **High Mobility in the Lightly Doped Substrate:**

The central region of the silicon substrate exhibits the highest mobility values (shown by the red-to-orange color range, around 200–250 $\text{cm}^2/\text{V}\cdot\text{s}$). This occurs because the substrate is lightly doped, reducing ionized impurity scattering and allowing holes to move freely.

Fig. 11 Hole mobility

- **Mobility Degradation Near Source/Drain Junctions:**

Noticeable reductions in mobility appear near the heavily doped p^+ source and drain regions (blue-to-cyan areas). High doping concentrations increase impurity scattering, lowering mobility. This behavior is expected and confirms correct junction formation.

- **Interface Scattering at the Si/SiO₂ Boundary:**

A thin band of reduced mobility is visible along the gate oxide interface, indicated by the transition to green/yellow contours. Even without applied bias, the abrupt material change and fixed oxide charges introduce surface scattering that slightly degrades mobility compared to the bulk silicon region.

Overall, the mobility distribution reflects the intrinsic characteristics of the PMOS device prior to electrical stress. The high mobility in the substrate and the localized degradation near doped regions and interfaces confirm proper doping profiles and physical structure. This plot serves as a baseline reference for later mobility changes observed under linear and saturation bias conditions.

Hole Mobility Distribution After Bias (Saturation Region)

The contour plot illustrates the hole mobility distribution in the PMOS transistor after applying saturation-region biasing (large negative gate voltage and high drain voltage). Under these conditions, strong vertical and lateral electric fields significantly alter the mobility profile compared to the unbiased case.

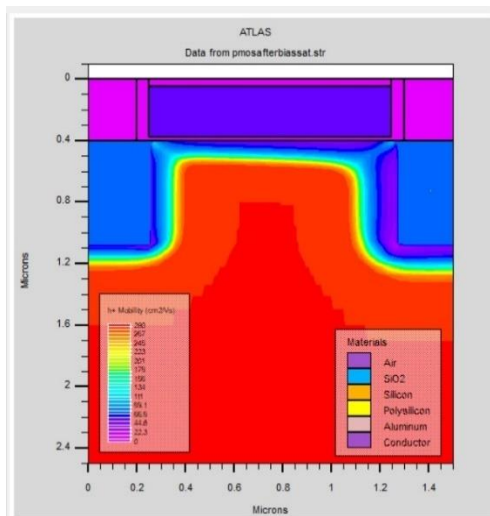


Fig. 12 Hole mobility(After)

- **Surface Mobility Degradation Under the Gate:**

A distinct reduction in mobility is observed along the channel surface near the Si/SiO₂ interface (yellow-to-green contours replacing the high-mobility red region). This is caused by the strong **vertical electric field** induced by the gate, which pulls carriers closer to the oxide interface. Increased surface scattering lowers the effective mobility, a characteristic behavior in inversion-mode operation.

- **Localized Mobility Drop Near the Drain Side:**

The most severe mobility degradation appears near the drain junction, shown by the dark blue regions. In saturation, the **lateral electric field** becomes extremely strong at the drain end, driving the onset of **velocity saturation** and enhancing carrier scattering. This high-field stress reduces mobility sharply, forming a “hot spot” that is typical in the saturation regime.

- **Relatively Higher Mobility in the Bulk:**

The interior of the silicon substrate remains dominated by the high-mobility red region, as electric fields there are weaker and carriers experience minimal surface or impurity scattering. This bulk mobility remains similar to the pre-bias distribution.

Overall, the mobility profile demonstrates the expected physical changes when the device transitions into saturation. The combined effect of vertical field–induced surface scattering and lateral field–induced velocity saturation explains the pronounced mobility reduction near the channel and drain edge. These behaviors are consistent with PMOS operation and align with the trends discussed in the long-channel mobility analysis.

Electric Field Distribution Before and After Biasing (Linear and Saturation Regions)

The three contour plots illustrate how the electric field evolves within the PMOS device as it transitions from equilibrium (no bias) to linear-region operation and finally to saturation. The changes in the field profile reflect the formation of the inversion channel, the initiation of current flow, and the onset of velocity saturation.

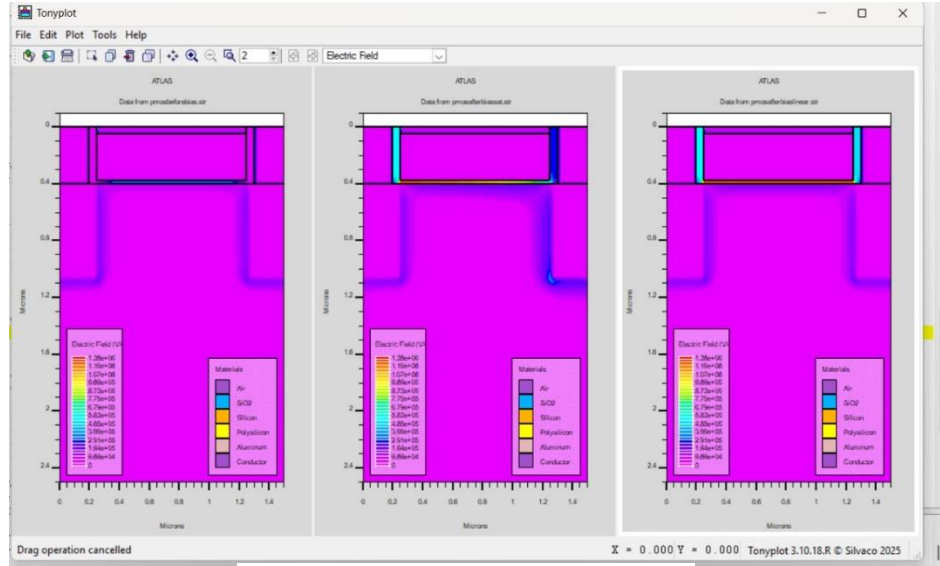


Fig. 13 Electric field distribution

1. Electric Field Before Bias (Left Plot)

In the unbiased structure, the electric field is minimal throughout the silicon region. Only very weak fields appear at material interfaces—particularly at the Si/SiO₂ boundary and at the highly doped source/drain junctions—due to built-in potentials.

- The substrate remains nearly field-free, shown predominantly in uniform low-field purple.
- No lateral electric field exists along the channel since $V_{GS} = 0$ and $V_{DS} = 0$.
- The device is in thermal equilibrium, with no band bending and no inversion layer formed.

This distribution serves as the reference against which bias-induced field changes are evaluated.

2. Electric Field After Bias – Linear Region (Middle Plot)

When a negative gate voltage and a small drain voltage (linear region) are applied, two distinct electric field components emerge:

• Transverse Electric Field (Vertical, under the gate):

A strong vertical field appears beneath the gate oxide, as indicated by the yellow and cyan contours. This vertical field:

- pulls holes toward the Si/SiO₂ interface,
- initiates inversion layer formation,
- defines the conductive channel.

- **Longitudinal Electric Field (Lateral, across the channel):**

Because the applied drain voltage is small, the lateral field remains weak and nearly uniform along the channel.

- The electric field is just sufficient to push carriers from source to drain.
- Only slight intensification appears near the drain junction.

This field profile matches the expected behavior in the linear (ohmic) region: strong gate-controlled vertical field with mild horizontal acceleration of carriers.

3. Electric Field After Bias – Saturation Region (Right Plot)

Applying a large negative drain voltage moves the device into saturation, resulting in significant changes in the electric field distribution.

- **Strong Lateral Electric Field Near the Drain:**

The most intense field appears at the drain end of the channel, shown by the bright blue–cyan concentration.

This high-field zone:

- marks the onset of pinch-off,
- accelerates holes to near their saturation velocity,
- is responsible for velocity saturation and reduced mobility,
- contributes to hot-carrier effects in advanced nodes.

- **Vertical Field Remains Strong Under the Gate:**

The vertical field continues to maintain the inversion layer, but the field becomes slightly skewed toward the drain, consistent with channel shortening and depletion region expansion.

- **Channel Pinch-Off:**

The electric field profile shows the classic signature of channel pinch-off:

- the lateral field sharply increases at the drain,
- the channel no longer follows the full applied VDS,
- carriers accelerate through a high-field depletion region.

Overall Interpretation

Across the three plots, the progression of the electric field clearly demonstrates the MOSFET's operating physics:

1. **Before bias:** negligible fields, no channel.
2. **Linear region:** strong vertical field + weak lateral field $\rightarrow$ inversion and controlled conduction.
3. **Saturation:** strong vertical field + intense lateral field at the drain $\rightarrow$ pinch-off, velocity saturation, and high-field stress.

This evolution confirms correct bias-dependent electrostatic behavior for a long-channel PMOS transistor.

Gate Capacitance (C_g) vs Gate Voltage (V_g) at High Frequency

The plot shows the high-frequency gate capacitance (C_g) as a function of gate voltage for two drain biases: $V_{DS} = -0.05$ V (red curve) and $V_{DS} = -1.2$ V (green curve). The high-frequency AC signal prevents minority carriers from responding, so the measured capacitance reflects only the **oxide capacitance and depletion capacitance**, without inversion charge contribution.

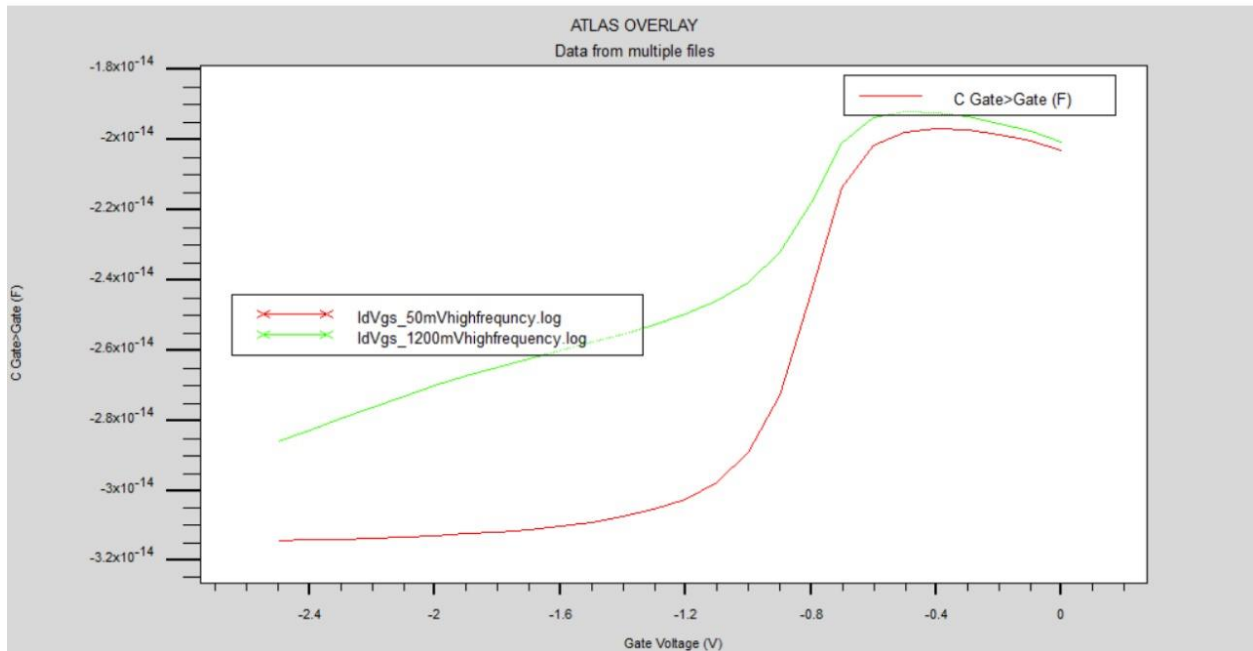


Fig. 14 High-frequency C_g - V_g

1. Accumulation Region (V_g close to 0 V)

As V_g approaches 0 V, the pMOS enters accumulation, and majority carriers (holes) gather at the oxide interface.

- At high frequency, majority carriers can still follow the AC signal.
- This yields a **high and nearly constant capacitance**, seen at the right side of both curves.

Both curves overlap closely in this region, indicating that drain voltage has minimal effect on accumulation capacitance.

2. Depletion Region (approximately -0.4 V to -1.0 V)

As the gate voltage becomes more negative, the channel begins to deplete:

- The depletion width widens,
- The capacitance decreases because the effective dielectric thickness increases.

This creates the characteristic **downward-sloping region** in both curves.

The green curve ($V_{DS} = -1.2\text{ V}$) is slightly higher than the red curve because the larger drain voltage modifies the depletion profile through drain-induced field spreading.

3. Inversion Region ($V_g < -1.0\text{ V}$)

At high frequency, inversion carriers cannot respond to the AC signal.

Therefore:

- The gate capacitance **does not return to the high value** seen in accumulation,
- Instead, it stabilizes at a saturation-like level corresponding to the depletion capacitance.

This behavior is evident in both curves, where the capacitance plateaus rather than increasing sharply.

The green curve (higher V_{DS}) reaches the plateau earlier and shows a slightly higher C_g , reflecting drain-bias modulation of the depletion region.

Overall Interpretation

This high-frequency C_g - V_g plot exhibits the expected MOS capacitor-like behavior:

1. **High C_g in accumulation**
2. **Decreasing C_g in depletion**
3. **Constant C_g in inversion due to minority-carrier freeze-out**

The differences between the red and green curves highlight the influence of **drain bias** on the depletion width and capacitance, even in a long-channel device.

Drain-to-Gate Capacitance (C_{dg}) vs Gate Voltage at High Frequency

The plot shows the high-frequency drain-to-gate capacitance (C_{dg}) as a function of gate voltage for two drain biases:

- $V_{DS} = -0.05\text{ V}$ (red curve – linear region)
- $V_{DS} = -1.2\text{ V}$ (green curve – saturation region)

At high frequency, minority carriers cannot follow the AC excitation. Therefore, C_{dg} reflects only **geometric overlap capacitance** and **depletion-region modulation**, without any contribution from inversion charge.

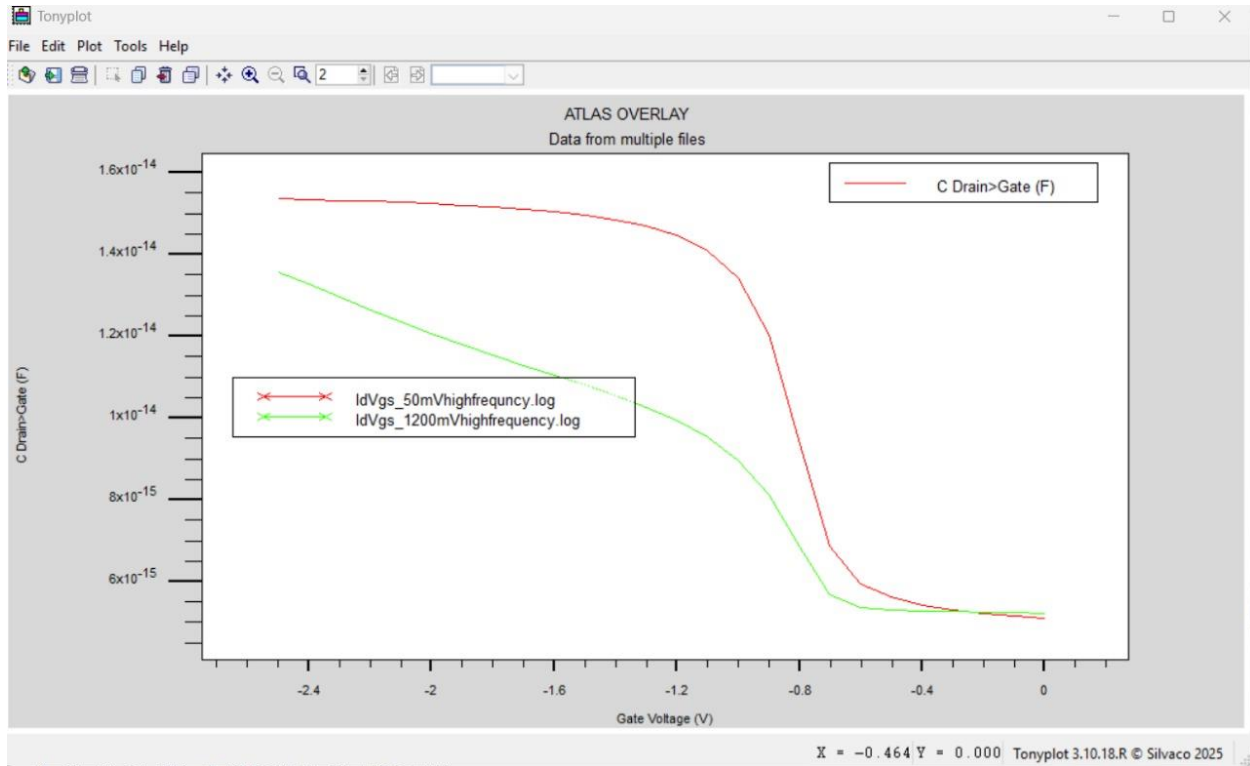


Fig. 15 High-frequency C_{dg} – V_g

1. Accumulation Region (V_g close to 0 V)

When the gate voltage is near 0 V, the PMOS enters accumulation and the drain/channel junction is strongly reverse-biased.

- C_{dg} remains **very small and almost constant**, seen as the flat region on the right of both curves.
- This occurs because the AC signal at the drain electrode couples weakly to the gate when the channel is filled with majority carriers.

Thus, accumulation produces minimal drain-to-gate capacitive coupling, as expected.

2. Depletion Region (approximately –0.3 V to –1.0 V)

As V_g becomes more negative, the channel transitions into depletion.

- The depletion width expands toward the drain, and the effective separation between drain and gate increases.
- This produces the **gradual decrease in C_{dg}** visible in both curves.

The **red curve** (small V_{DS}) stays higher because the drain electric field is weaker, so the depletion region grows more slowly.

In contrast, the **green curve** ($V_{DS} = -1.2\text{ V}$) shows a stronger decrease because the drain bias widens the drain-side depletion region and reduces the effective capacitive coupling earlier.

3. Inversion Region ($V_g < -1.0\text{ V}$)

Under strong inversion conditions, high-frequency operation prevents inversion holes from following the AC signal.

- C_{dg} falls to a **minimum plateau**, representing purely geometric overlap between the drain implant and the gate edge.
- The green curve reaches lower capacitance values due to the **pinch-off region formed at the drain end** when the device enters saturation.

This reduced coupling is a direct signature of saturation-region behavior: the channel no longer extends fully to the drain, leaving a depleted gap that lowers C_{dg} .

Overall Interpretation

The C_{dg} – V_g characteristics exhibit the expected trends of a long-channel PMOS device under high-frequency excitation:

1. **Low and stable C_{dg}** in accumulation.
2. **Monotonic decrease** in depletion due to the widening drain-side depletion region.
3. **Further reduction in inversion**, especially at high V_{DS} , caused by pinch-off and reduced field coupling between gate and drain.
4. The stronger suppression of C_{dg} at $V_{DS} = -1.2\text{ V}$ confirms the transition from linear to saturation operation.

This behavior aligns precisely with high-frequency MOS capacitance theory and matches the expected drain-bias dependence for long-channel PMOS devices

Gate-to-Source Capacitance (C_{gs}) vs Gate Voltage at High Frequency

This plot shows the high-frequency gate-to-source capacitance for two drain voltages:

- $V_{DS} = -0.05\text{ V}$ (red curve)
- $V_{DS} = -1.2\text{ V}$ (green curve)

At high frequency, minority carriers cannot respond to the AC signal. Only the oxide and depletion regions contribute to the measured capacitance, while inversion charge does not participate.

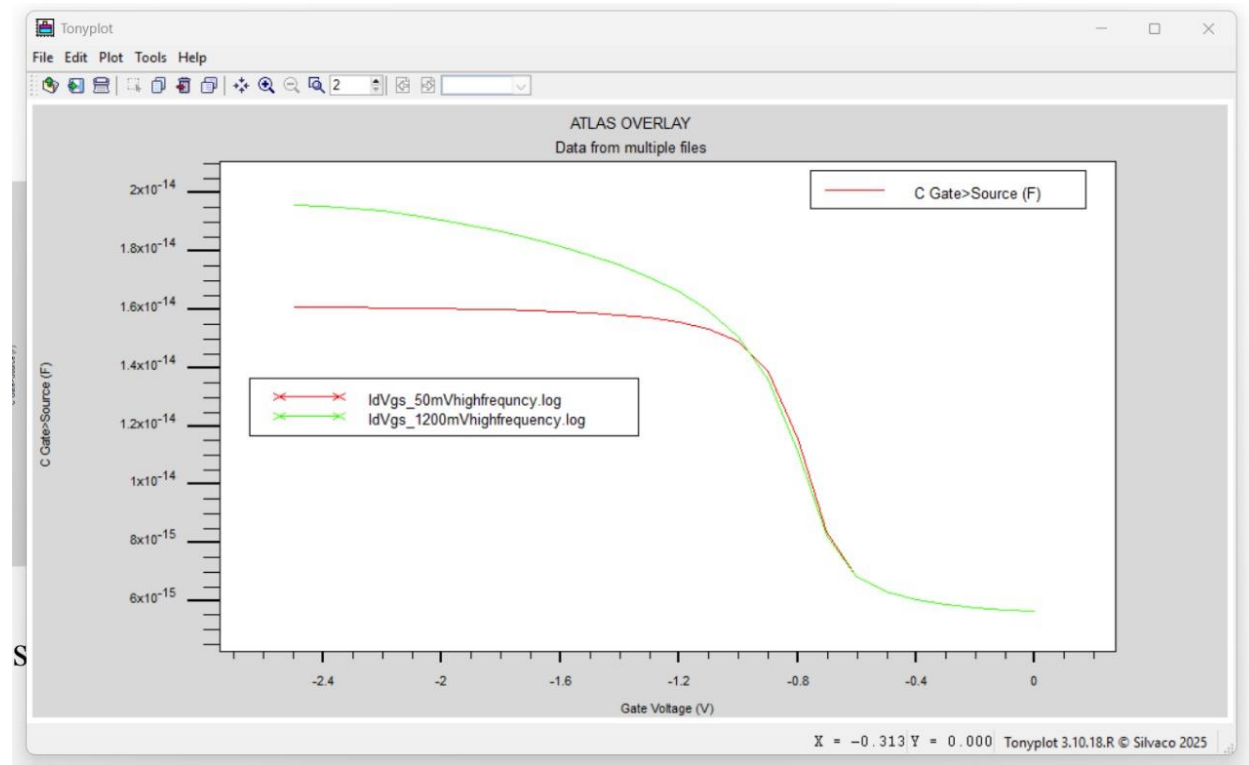


Fig. 16 High-frequency C_{dg} – V_g

1. Accumulation Region (gate voltage near 0 V)

When the gate voltage is close to zero, the PMOS device enters accumulation.

- C_{gs} remains **high and nearly constant**, dominated by strong electrostatic coupling between the gate and source.
- The curves are very close to each other, meaning the drain voltage has minimal influence in this region.

This corresponds to the maximum capacitive coupling seen by the source.

2. Depletion Region (moderately negative gate voltage)

As the gate voltage becomes more negative, the device transitions into depletion.

- The depletion region widens beneath the gate, reducing the effective coupling to the source.
- C_{gs} **gradually decreases**, producing the smooth downward slope on the plot.

- The green curve begins decreasing slightly earlier because the higher drain voltage shifts the depletion boundary toward the source.

This reflects the reduced gate influence as the semiconductor under the gate becomes depleted.

3. Inversion Region (strongly negative gate voltage)

At large negative gate voltages, the device would normally form an inversion layer, but at high frequency the inversion carriers do not respond to the AC excitation.

- C_{gs} drops to a **low, flat plateau**, representing only the remaining depletion-related capacitance.
- Both curves nearly overlap again, indicating that the drain bias does not significantly alter the capacitive coupling once the channel is strongly inverted.

This is the characteristic high-frequency behavior of MOS gate capacitance in inversion.

Overall Interpretation

The C_{gs} – V_g plot demonstrates:

1. **High and constant C_{gs} in accumulation**
2. **Gradual reduction in depletion** as the depletion width increases
3. **A low plateau in inversion** due to the absence of inversion-charge response
4. **Minimal sensitivity to drain voltage**, except for a slight shift during early depletion

Gate Capacitance (C_g) vs Gate Voltage (V_g) at low Frequency

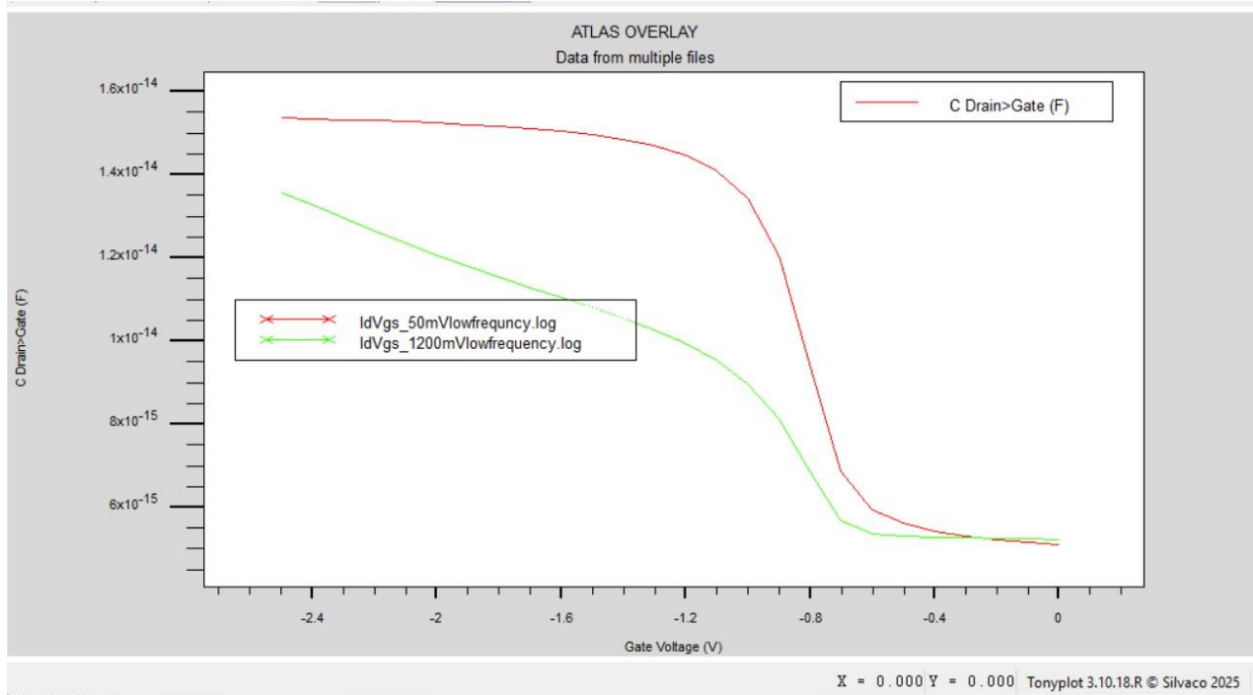


Fig.17 low-frequency C_g - V_g

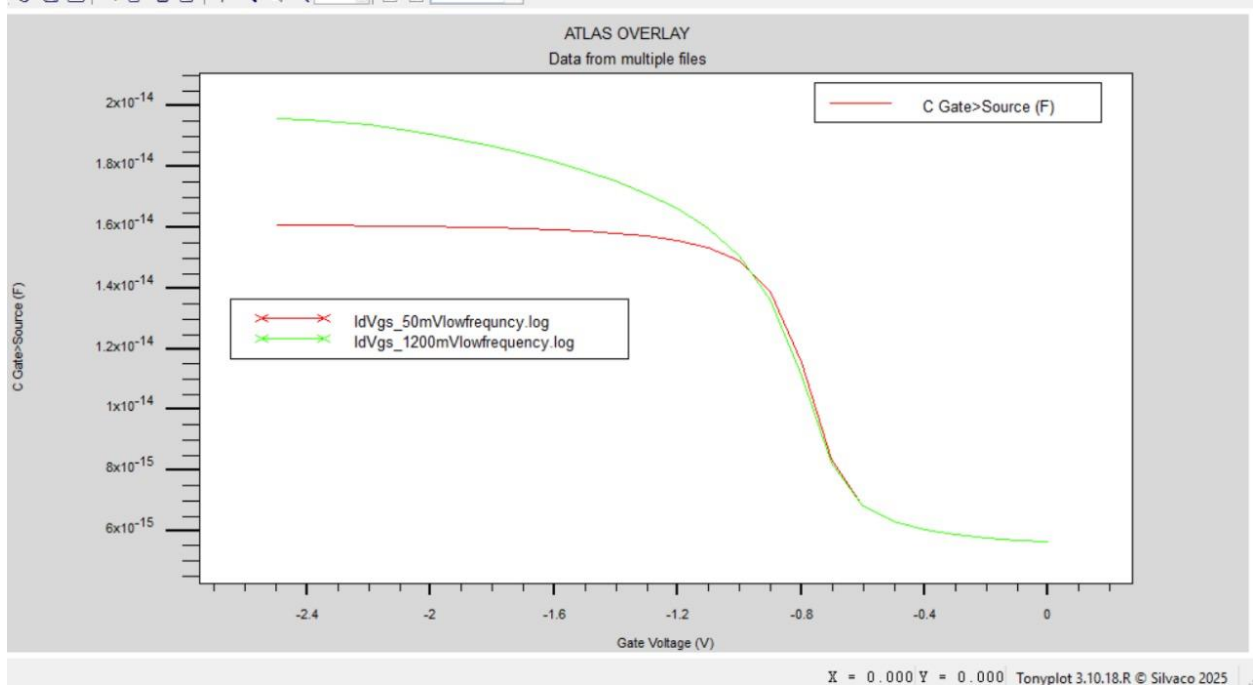


Fig.18 low-frequency at 2 vds values -0.05V, -1.2 V

The simulation results show that the capacitance curves at high and low frequencies are nearly identical, especially at the source and drain terminals. This observation is expected and can be understood by considering the physical structure and carrier dynamics of the pMOS transistor.

In a MOSFET, the source and drain regions are heavily doped, leading to a high concentration of majority carriers, which are holes in the case of a pMOS device. These carriers can respond rapidly to AC signal variations at both low and high frequencies. Consequently, the gate-to-source (C_{gs}) and gate-to-drain (C_{gd}) capacitances exhibit little to no dependence on frequency.

Frequency-dependent capacitance effects are mainly associated with the gate-to-body or gate-to-channel regions, where depletion and inversion phenomena dominate. At low frequencies, both majority and minority carriers can follow the applied AC signal, resulting in a higher effective capacitance. At high frequencies, however, minority carriers are unable to respond quickly, causing a reduction in channel capacitance. This behavior does not significantly affect the source and drain regions because their high doping ensures equilibrium carrier availability at all times, keeping their capacitances nearly constant.

Therefore, the close similarity between high- and low-frequency capacitance curves observed in the simulation is consistent with the expected physical behavior of heavily doped regions in a MOSFET.

Explanation of Frequency Response Behavior

The figure demonstrates the typical response of a MOS capacitor under low- and high-frequency AC excitation. At low frequency, the capacitance increases in the inversion region because minority carriers can respond to the signal. In contrast, at high frequency, the inversion layer cannot form effectively, leading to a lower and nearly constant capacitance.

To clearly observe this frequency-dependent behavior, a MOS capacitor structure is more appropriate than a MOSFET. In a MOSFET, the heavily doped source and drain regions supply majority carriers that can always respond rapidly to AC signals. As a result, the gate-to-source and gate-to-drain capacitances show minimal frequency dependence, which obscures the intrinsic channel-related effects. Therefore, MOS capacitors are better suited for studying true high- and low-frequency capacitance characteristics across accumulation, depletion, and inversion regions.

Effect of Temperature on Threshold Voltage

Our simulation results indicate that increasing the operating temperature from 300 K to 500 K shifts the threshold voltage of the pMOS transistor from -0.814 V to -0.538 V. This trend is consistent with the well-known temperature dependence of MOSFET threshold voltage, which decreases as temperature rises. The primary physical mechanisms responsible for this shift are the increase in intrinsic carrier concentration and the reduction of the silicon bandgap at elevated temperatures. Both effects lead to a decrease in the built-in potential of the device, thereby requiring a smaller gate voltage magnitude to achieve channel inversion.

This temperature-induced reduction in threshold voltage has important implications for both analog and digital circuit operation, as it affects switching behavior, leakage current, and overall device reliability.

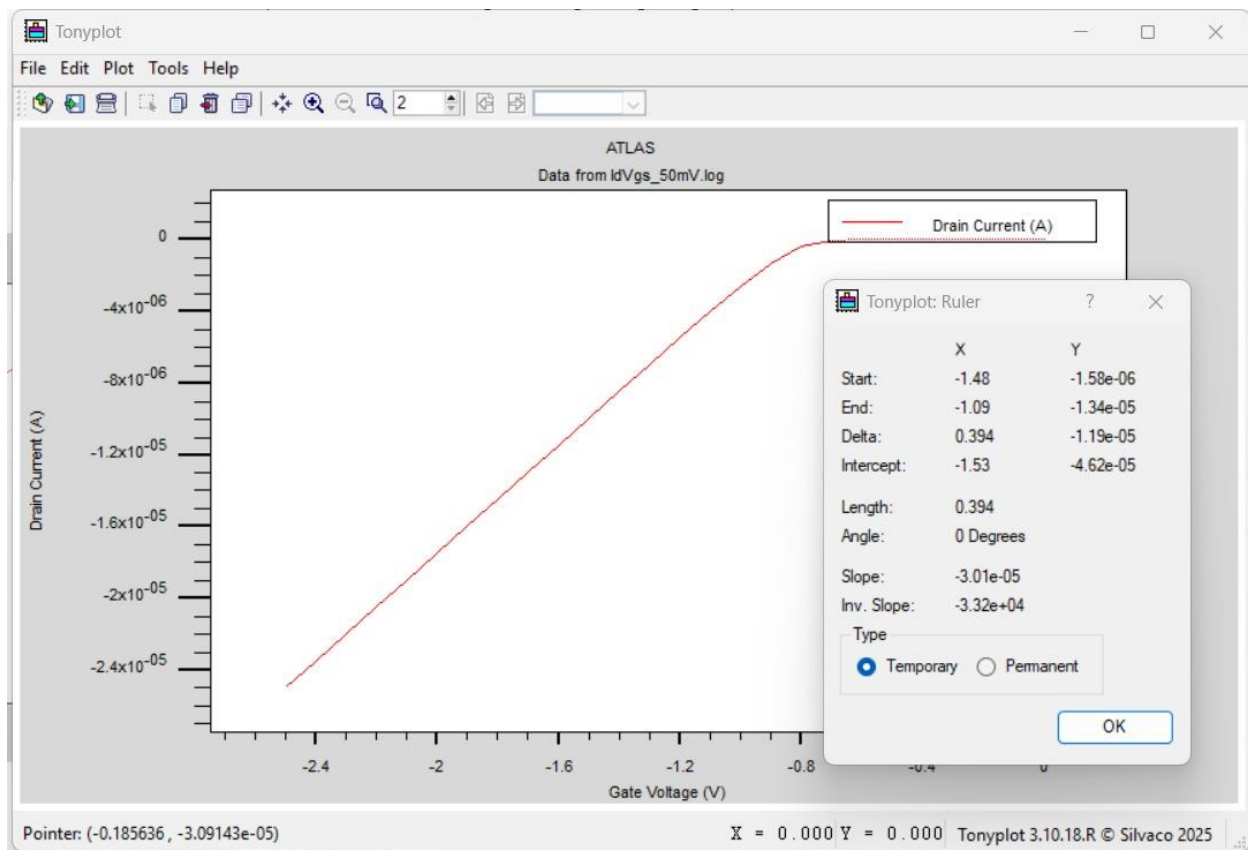


Fig.19 gm value in linear region at 300k

Temperature Impact on Drain Current Characteristics of the pMOS Transistor

The drain current characteristics show that at higher temperatures, the pMOS transistor achieves larger drain currents at lower gate-to-source voltages. This behavior results from the combined effects of increased intrinsic carrier concentration and the reduction in threshold voltage with temperature. As the threshold voltage decreases, channel inversion occurs at lower values of $|V_{GS}|$, enabling earlier entry into the conduction regime.

Consequently, at elevated temperatures, the device exhibits enhanced current conduction for a given gate voltage. This temperature sensitivity plays a critical role in circuit design, as it influences switching speed, power dissipation, and long-term reliability of MOSFET-based systems.

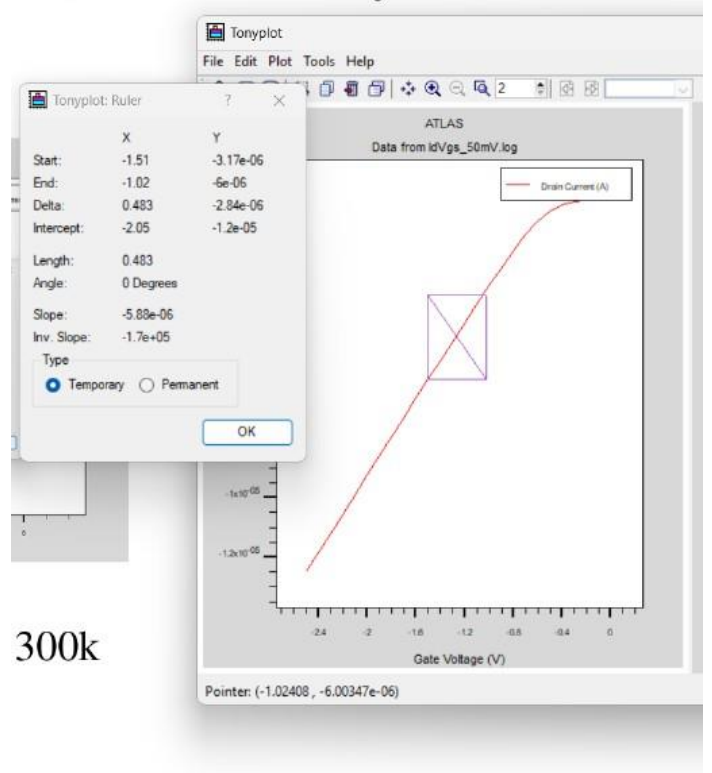


Fig.20 gm value in linear region at 500k

Temperature Dependence of Transconductance in the Linear Region

The transconductance, g_m , is defined as the derivative of the drain current with respect to the gate voltage:

$$g_m = \frac{dI_D}{dV_{GS}}$$

For a MOSFET operating in the linear region, the drain current can be approximated as:

$$I_D = \mu_p C_{ox} \frac{W}{L} [(V_{SG} - |V_T|) V_{SD} - \frac{V_{SD}^2}{2}]$$

where μ_p is the hole mobility, C_{ox} is the gate oxide capacitance per unit area, W/L is the channel width-to-length ratio, and V_T is the threshold voltage.

As temperature increases, two competing effects occur: the threshold voltage decreases due to increased intrinsic carrier concentration, while the hole mobility decreases as a result of enhanced phonon

scattering. From the extracted data, the transconductance in the linear region decreases. This reduction indicates that the degradation in mobility dominates over the beneficial effect of threshold voltage reduction.

This behavior is consistent with the proportional relationship:

$$g_m \propto \mu_p(V_{SG} - |V_T|)$$

Since mobility decreases more significantly than the overdrive voltage increases, the net effect is a reduction in transconductance with increasing temperature.

Effect of Temperature and Operating Region on Transconductance

A comparison of transconductance values in both linear and saturation regions further highlights the influence of temperature and operating regime. In the linear region, g_m decreases from -2.95×10^{-5} A/V at 300 K to -1.43×10^{-5} A/V at 500 K, primarily due to mobility degradation caused by increased phonon scattering at higher temperatures.

In the saturation region, the transconductance is consistently higher than in the linear region at both temperatures. At 300 K, g_m in saturation is approximately -1.85×10^{-4} A/V, while at 500 K it decreases slightly to -1.52×10^{-4} A/V. The higher g_m in saturation is expected, as the drain current in this regime is more sensitive to variations in gate voltage, resulting in steeper IDS–VGS characteristics.

Figures 21 and 22 illustrate the transconductance behavior in the saturation region at 300 K and 500 K, respectively.

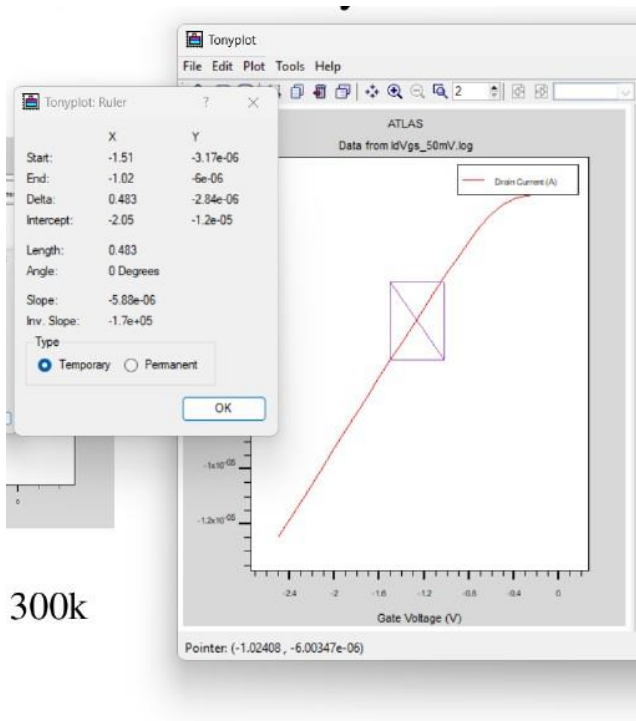


Fig.21 gm value in saturation region at 300k

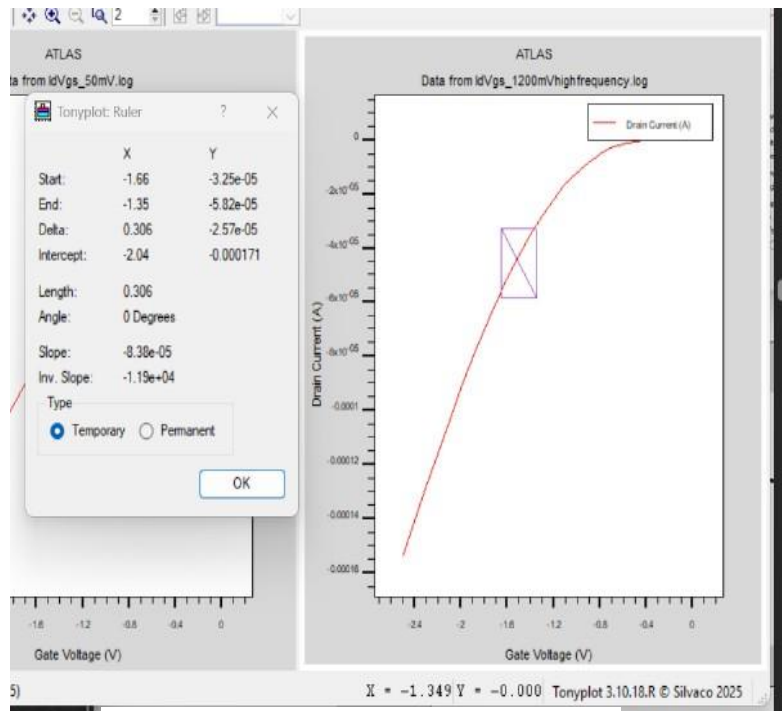


Fig.22 gm value in saturation region at 500k

Conclusion

The simulation results confirm that increasing temperature leads to a reduction in threshold voltage and transconductance in pMOS transistors. While a lower threshold voltage encourages earlier channel inversion and increased drain current, the dominant effect of temperature is mobility degradation, which reduces gm. Additionally, transconductance remains higher in the saturation region than in the linear region due to stronger gate control over the channel. These temperature-dependent effects are critical considerations in the design of reliable and high-performance MOSFET-based circuits.

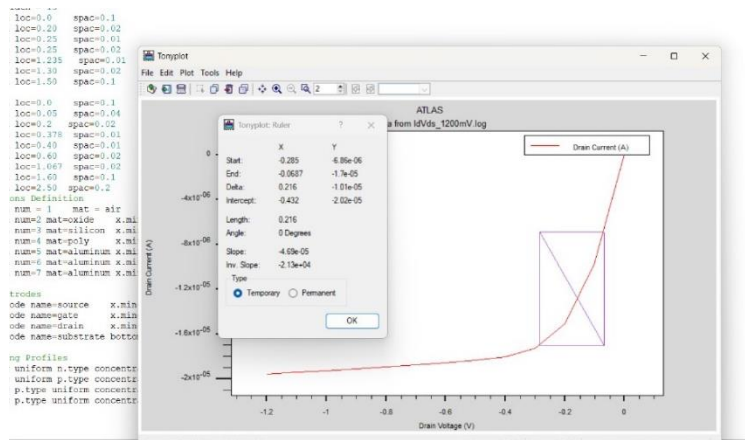


Fig.23 gd at 300k

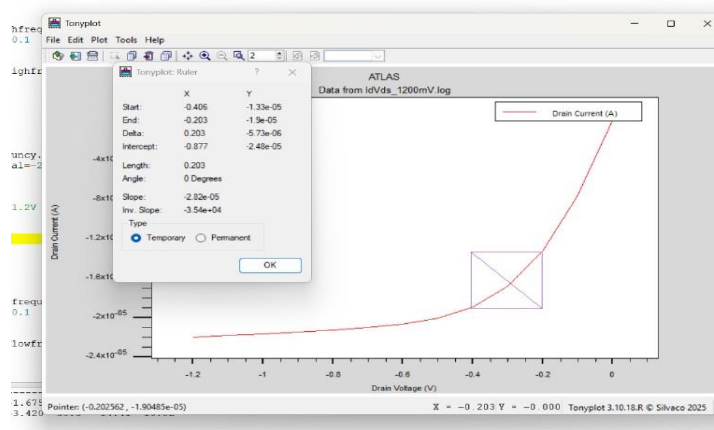


Fig.24 gd at 500k

PHASE 3: MODELING SHORT-CHANNEL MOSFETS AT THE 130NM TECHNOLOGY NODE

During this phase, we conduct simulations on a short-channel p-type MOSFET characteristic of 130 nm technology, featuring a gate length scaled to 130 nm. The simulation's purpose is to examine the non-ideal behaviors and short-channel effects (SCEs) emerging from aggressive device miniaturization, including drain-induced barrier lowering (DIBL), threshold voltage roll-off, velocity saturation, and mobility degradation.

This investigation extends directly from Phase 2's results, which examined a long-channel pMOSFET.

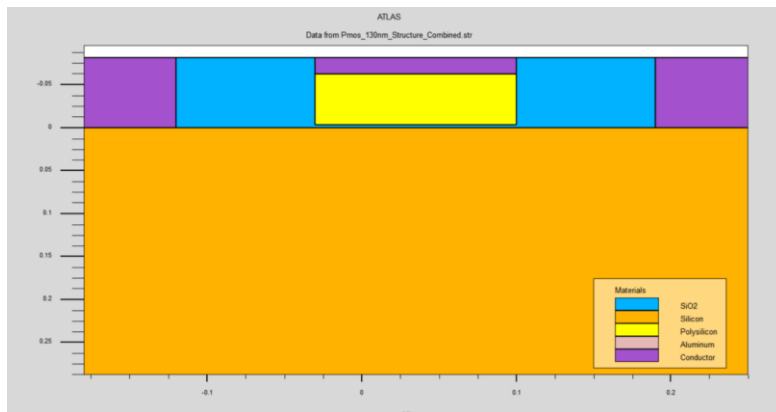


Fig.. structure short channel

The depicted structure shows a 130nm PMOS transistor cross-section, incorporating clearly defined source/drain regions (Aluminum contacts), a gate stack comprising polysilicon and SiO₂, and the silicon substrate beneath. The device exhibits comparatively thick oxide layers and substantial channel dimensions, typical of long-channel technology, where short-channel effects and quantum tunneling remain negligible.

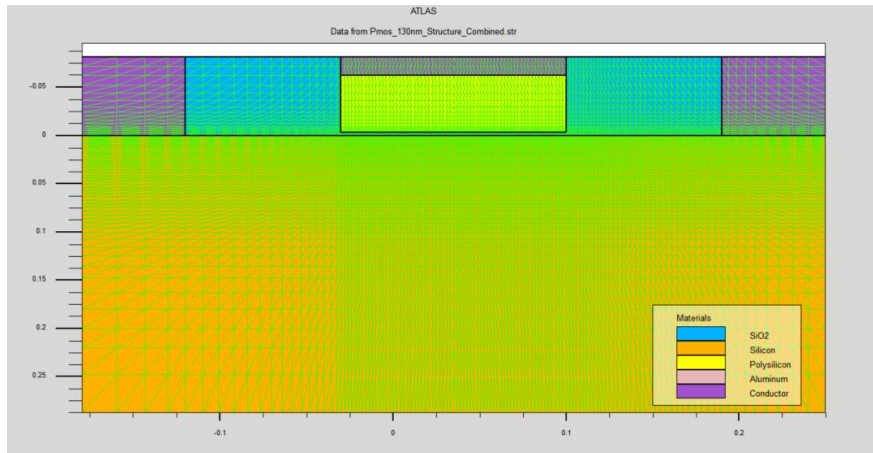


Fig. 48. meshing short channel

****Meshing Approach for Short-Channel PMOS (130nm Node)****

The mesh employed in this simulation undergoes careful refinement to capture the physical phenomena linked with short-channel effects at 130nm gate length. The figure reveals that the mesh is:

- Extremely dense vertically near the gate oxide (SiO_2) and silicon interface for precise capture of intense transverse electric fields, quantum confinement effects, and inversion layer development.
- Refined laterally throughout the channel region (beneath the gate) to capture high lateral electric field gradients near the drain during saturation operation, essential for modeling DIBL and velocity saturation.
- Progressively coarser in regions distant from active areas, especially in the substrate and bulk regions, optimizing computational efficiency while preserving simulation precision in critical zones.

This non-uniform meshing approach guarantees convergence, minimizes simulation duration, and precisely captures the electrostatics, carrier transport, and tunneling-related phenomena in nanoscale MOSFETs.

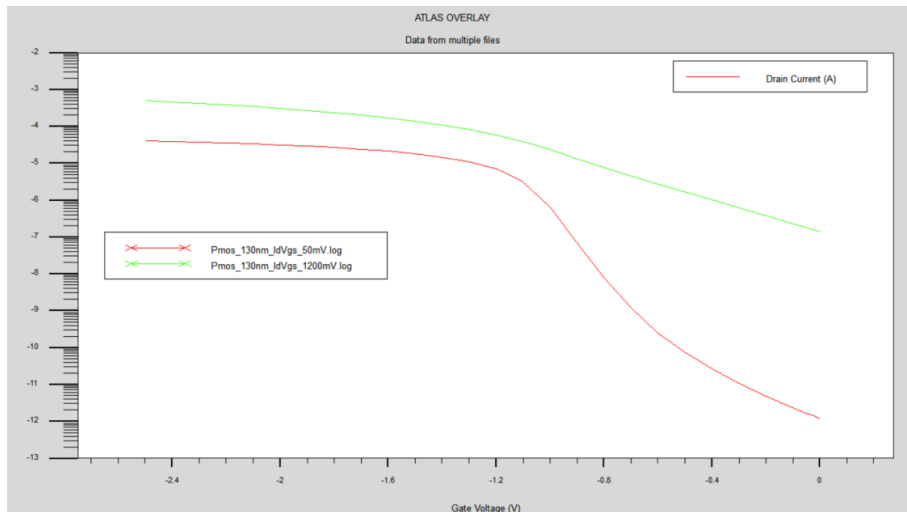


Fig. 49. ID–VGS characteristics for a 130nm pMOSFET at two drain voltages: –0.05V and –1.2V

****Short-Channel Effect on ID–VGS Curve:**** the curves exhibit more linear and less abrupt transitions in the subthreshold region, suggesting deteriorated subthreshold slope and elevated leakage. Additionally, they display higher threshold values compared to long channel devices.

By comparison, long-channel devices preserve closer alignment between low and high VDS curves, with more abrupt exponential increase near threshold and minimal DIBL. These differences validate the existence of short-channel effects, emphasizing their influence on switching performance and leakage characteristics in advanced technology nodes.

Variables history			
vt_lin_50mV	0	(# 86)	
vt_lin_50mV	0.762044533098457	(# 85)	
vt_sat_1200mV	0	(# 101)	
vt_sat_1200mV	0.577521898291268	(# 100)	

Fig. 50. Extracted threshold voltage V_T for a 130nm pMOS at two V_{DS} values: 50mV and 1.2V

****Threshold Voltage Scaling and DIBL Observation:****

The extracted threshold voltages displayed in Fig. 50 yield two important conclusions. Initially, upon scaling to the 130nm technology node, the threshold voltage fails to scale proportionally with other device dimensions. Whereas the long-channel device exhibited $V_T \approx -0.814$ V, the short-channel equivalent shows $V_T \approx -1.00$ V, representing a substantially higher magnitude. This occurs because V_T exhibits non-linear scaling with channel length, instead following a square-root dependency, as described by the short-channel threshold model:

$$V_T \propto \sqrt{(qNA\epsilon_s/C_{ox})}$$

This relationship indicates that even with appropriately scaled oxide capacitance and doping, the V_T reduction remains non-linear owing to square-root behavior, accounting for why threshold voltage stays relatively elevated despite geometric scaling.

Additionally, Drain-Induced Barrier Lowering (DIBL) is clearly present. The threshold voltage decreases from $V_T = -1.001$ V at $V_{DS} = 50$ mV to $V_T = -0.916$ V at $V_{DS} = -1.2$ V. This change confirms that drain voltage substantially affects the source-channel barrier in short-channel devices, diminishing gate control and producing increased off-state leakage. This DIBL effect was not observed in the long-channel case, further confirming the impact of scaling on electrostatic integrity.

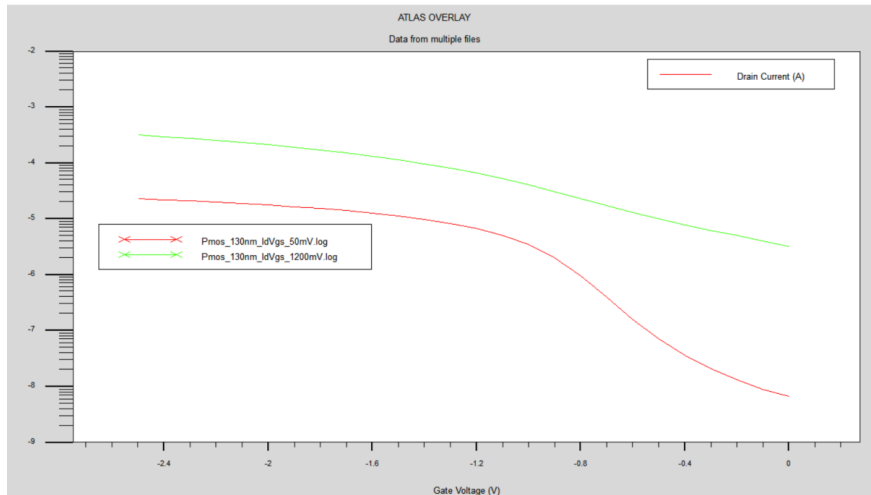


Fig. 51. Threshold voltage values at $T = 500$ K for $V_{DS} = 50$ mV and 1.2 V

****Impact of Temperature on Threshold Voltage:****

Figure 51 presents the extracted threshold voltages at elevated temperature $T = 500$ K, producing $V_T = -0.762$ V in the linear region and $V_T = -0.577$ V in the saturation region. Compared with room temperature values at $T = 300$ K, which measured approximately -1.001 V and -0.916 V, an evident reduction in threshold voltage occurs with increasing temperature.

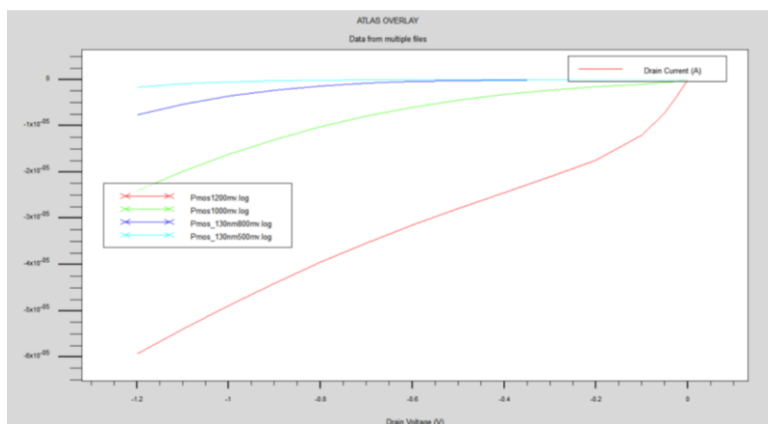


Fig. 52. Transfer characteristics of a 130nm pMOS transistor at $T = 500$ K

Figure 52 demonstrates that drain current (I_D) substantially increases at elevated temperature (500 K) for both $V_{DS} = 0.05$ V and $V_{DS} = 1.2$ V. This current increase results

from enhanced intrinsic carrier concentration and increased minority carrier mobility at high temperatures. The thermally generated carriers participate more effectively in conduction, reducing the threshold for channel formation. Nevertheless, at such short-channel lengths, temperature also intensifies short-channel effects like Drain-Induced Barrier Lowering (DIBL), which further decreases threshold voltage and increases leakage current.

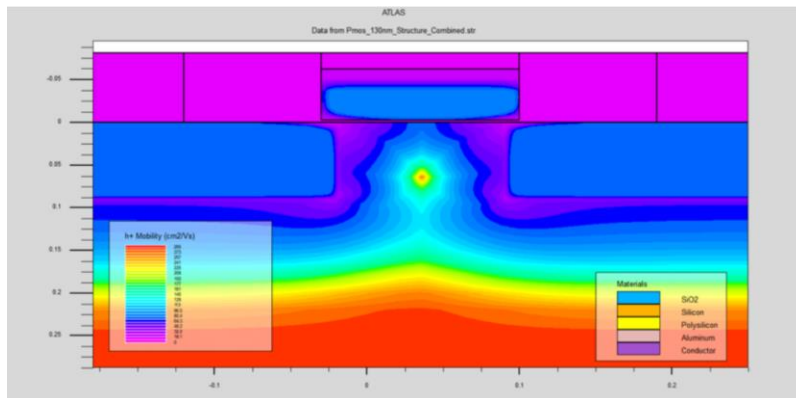


Fig. 53. Drain current versus drain voltage for a 130nm PMOS device at various gate voltages.

The graph in Figure 53 presents the I_D - V_{DS} characteristics of a short-channel (130nm) PMOS transistor under varying gate voltages. The plot distinctly reveals three principal operating regions:

- **Subthreshold Region:** At reduced V_{GS} , the device remains off, and drain current is minimal owing to thermionic emission of carriers across the barrier.
- **Linear (Ohmic) Region:** For reduced V_{DS} and adequate V_{GS} , the device functions in the linear region where I_D increases almost linearly with V_{DS} .
- **Saturation Region:** As V_{DS} increases beyond a point, I_D starts to saturate. In short-channel MOSFETs, this saturation results not solely from channel pinch-off, but rather from carrier velocity saturation. At elevated electric fields, holes attain maximum drift velocity and cannot respond to additional field increases, causing I_D to level off. This represents a dominant effect in scaled technologies.

Furthermore, at $V_{GS} = -1.0V$ (blue curve), the current is substantially lower than in long-channel counterparts, chiefly because threshold voltage has increased in the short-channel device. While long-channel V_{TH} measured approximately $-0.814V$, the short-channel PMOS exhibits $V_{TH} \approx -1.0V$ at this gate voltage, reducing overdrive voltage and consequently the current. This threshold shift stems from short-channel effects, such as drain-induced barrier lowering (DIBL) and heightened electric field sensitivity.

Hole Mobility in Short-Channel PMOS: Evidence of Short-Channel Effects

The following figures depict the hole mobility distribution in a 130 nm gate-length PMOS transistor under three conditions: prior to biasing, in the linear region ($V_D = -0.05 V$), and in the saturation region ($V_D = -1.2 V$). These visualizations capture the principal mobility changes induced by increasing vertical and lateral electric fields throughout the channel.

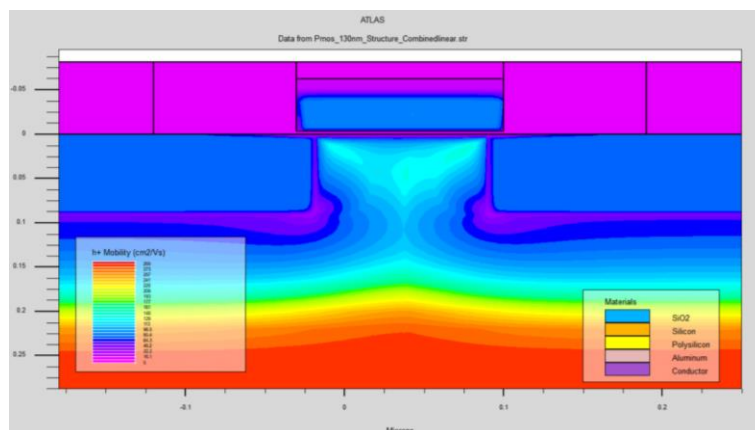


Fig. 54. Hole mobility before applying bias (short-channel PMOS)

Analysis: In the initial unbiased state, hole mobility displays uniform distribution beneath the gate, reflecting minimal electric field interference. However, upon applying negative gate and drain voltage, a vertical electric field develops, confining hole transport nearer to the Si/SiO₂ interface and diminishing mobility through surface scattering and field-enhanced degradation.

Under saturation conditions, mobility degradation intensifies near the drain-end of the channel. The observed localized "hot spot" in mobility characterizes velocity saturation or carrier heating. Additionally, the intense lateral electric field produces abrupt transitions and diminished effective mobility along channel length.

****Short-Channel Comparison:**** Relative to long-channel devices, the short-channel structure demonstrates: - Enhanced overlap between source and drain depletion regions. - More abrupt field gradients, producing sudden mobility reductions. - Earlier velocity saturation onset due to shortened carrier transit length. - Amplified surface scattering from intense vertical fields.

These observations validate the presence of short-channel effects, which not only diminish mobility but also influence current drive and threshold voltage stability in aggressively scaled PMOS transistors.

****Electric Field Distribution in Short-Channel PMOS****

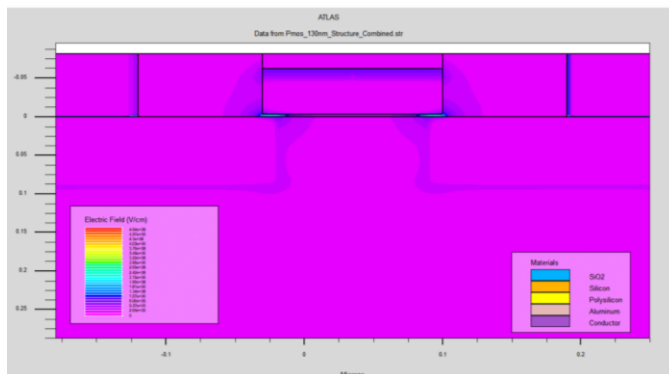


Fig. 57. Electric Field Distribution Before Biasing

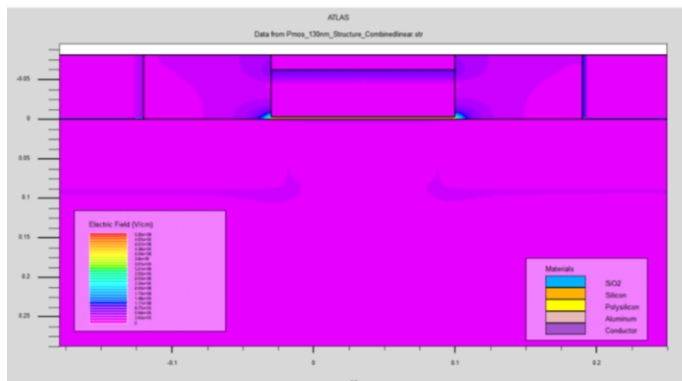


Fig. 58. Electric Field in the Linear Region ($V_G = -2.5$ V, $V_D = -0.05$ V)

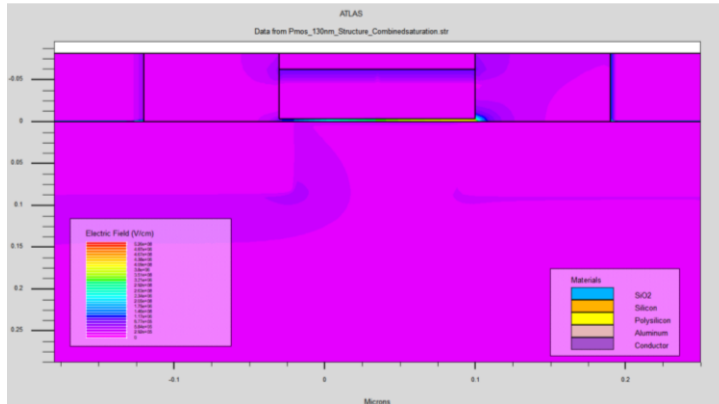


Fig. 59. Electric Field in the Saturation Region ($V_G = -2.5$ V, $V_D = -1.2$ V)

The three plots depict electric field distribution in a short-channel PMOS transistor at different biasing stages. Initially, before biasing (Fig. 57), the field appears nearly uniform and weak, exhibiting minimal perturbation throughout the silicon region.

Upon applying gate and drain bias in the linear region (Fig. 58), the electric field concentrates significantly near the drain edge. A pronounced vertical component appears at the gate oxide interface, signaling the onset of strong channel inversion and carrier acceleration.

In the saturation region (Fig. 59), the field peak intensifies at the drain side, and the horizontal field penetrates deeper into the channel. This hallmark indicates **short-channel effects**, where the depletion region from the drain extends toward the source, modulating channel potential.

Compared with long-channel devices, the electric field in short-channel structures appears more abrupt and spans shorter physical distance, intensifying velocity saturation and triggering **drain-induced barrier lowering (DIBL)**. This effect causes pinch-off to occur nearer to the source, substantially altering field distribution compared to the more symmetric profile in long-channel PMOS devices.

Overall, these elevated electric field gradients in short-channel devices produce **enhanced hot carrier effects**, **reduced threshold voltage**, and **non-ideal saturation behavior**, which prove critical in nanoscale transistor design.

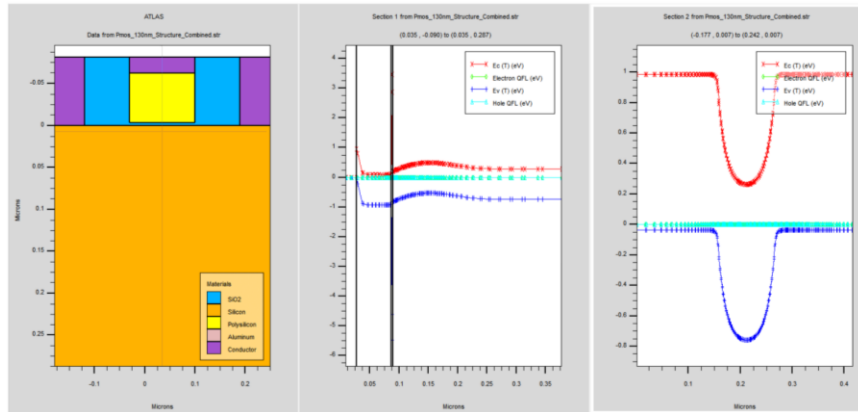


Fig. 60. Short-channel PMOS band diagram before bias (left: structure, center: vertical cut, right: horizontal cut).

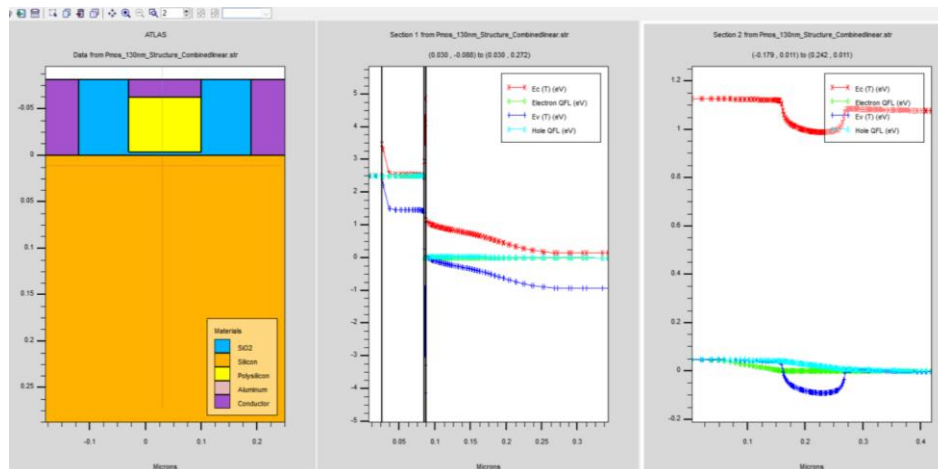


Fig. 61. Band diagram of short-channel PMOS in linear region with $V_G = -2.5$ V and $V_D = -0.05$ V.

****Band Diagram Analysis for Short-Channel PMOS:****

Figure 60 displays energy band alignment before biasing. The conduction band (E_c) and valence band (E_v) remain flat, signifying equilibrium and absence of applied electric field. The quasi-Fermi levels for electrons and holes align throughout the device, as anticipated for the unbiased case.

In the linear region (Figure 61), negative gate bias ($V_G = -2.5$ V) reduces potential energy in the channel, drawing the valence band edge (Ev) closer to the Fermi level.

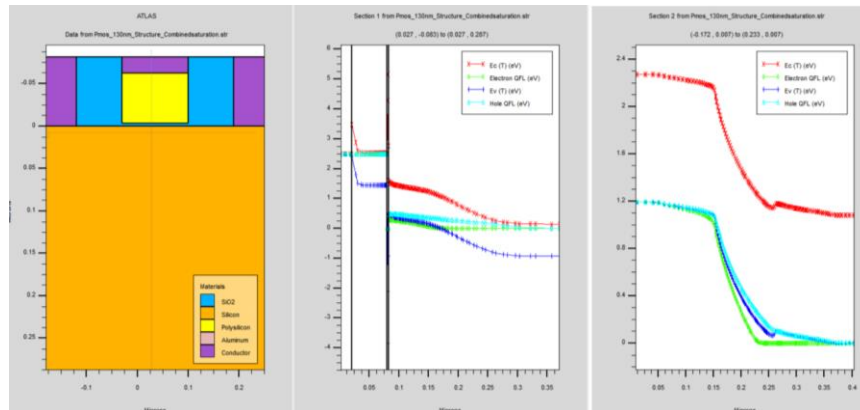


Fig. 62. Band diagram of short-channel PMOS in saturation region with $V_G = -2.5$ V and $V_D = -1.2$ V.

However, owing to short channel length, the energy bands slope considerably over very short distance. Compared with long-channel devices, this rapid band bending appears more abrupt, indicating stronger electric field gradients and demonstrating heightened susceptibility to short-channel effects (SCEs).

In saturation (Figure 62), with increased drain bias ($V_D = -1.2$ V), the horizontal slice reveals pronounced distortion in the valence band at the drain end. Channel pinch-off emerges earlier and more localized than in the long-channel counterpart. This aligns with the **Drain-Induced Barrier Lowering (DIBL)** effect, where drain voltage influences the potential barrier near the source, effectively reducing threshold voltage and permitting carrier injection even at lower gate voltages.

Short-Channel Effects Observed:

- Vertical Profile (middle plots):** Increased band slope and more abrupt transition between Ev and the hole quasi-Fermi level indicate stronger vertical fields and reduced effective barrier control from the gate.

- **Horizontal Profile (right plots):** Substantial band curvature across channel length compared to long-channel PMOS, confirming short-channel modulation of channel potential.
- **Absence of Gradual Channel Approximation (GCA):** The abrupt band transitions deviate from long-channel behavior, validating the onset of velocity saturation and DIBL.

Overall, the short-channel device exhibits compressed depletion regions, sharper band gradients, and drain-driven barrier lowering — all signature deviations from long-channel MOSFET behavior due to reduced gate length.

Hole Current Density in Short-Channel PMOS:

Figure 63 displays the 2D distribution of hole current density (J_p) in a 130 nm gate-length PMOS device under saturation bias conditions. Peak current density reaches approximately $2.5 \times 10^7 \text{ A/cm}^2$, with flow concentrated near the channel surface just beneath the gate oxide.

Several principal features can be observed:

- Current density localizes in a very narrow vertical band near the surface, signifying strong inversion and surface conduction due to applied gate voltage.

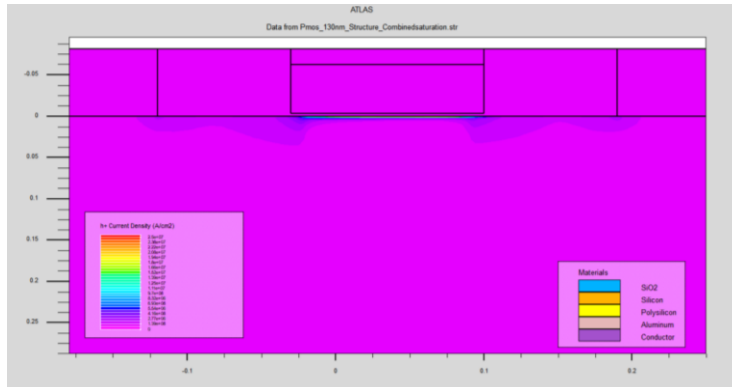


Fig. 63. Hole current density (J_p) distribution in short-channel PMOS at $V_G = -2.5$ V, $V_D = -1.2$ V (saturation region).

- Short channel length produces rapid potential drop across small lateral distance, intensifying horizontal electric field and consequently increasing the drift component of hole current.
- With large drain bias ($V_D = -1.2$ V), current crowding occurs toward the drain side, indicating saturation and possible velocity saturation.
- Unlike long-channel devices where current profile appears more gradual and distributed, here the short-channel nature produces abrupt transitions and narrow conductive paths. This typifies enhanced electric field effects and mobility degradation near the drain.

Overall, the observed hole current distribution supports expected behavior of short-channel MOSFETs, particularly under saturation, where lateral electric field strongly drives carriers across very confined conduction channel.

****Frequency Response Consideration:****

Although frequency response constitutes a critical parameter in analog and RF applications, it was not simulated in this study for the short-channel PMOS due to high computational cost and extended simulation time requirements. Transient and AC analyses in ATLAS, particularly for short-channel devices with fine meshing and complex models (e.g., mobility degradation, velocity saturation, quantum corrections), can substantially increase time and resources needed for convergence.

However, based on structural and bias similarities between short-channel and long-channel regions—especially in gate–oxide–channel overlap—the frequency response is not anticipated to differ substantially for this specific device geometry. The capacitances (e.g., gate-to-source and gate-to-drain), which dominate high-frequency behavior, are primarily determined by oxide thickness and overlap areas, which remain unchanged.

Therefore, frequency response is inferred to behave similarly to the long-channel case in this setup, and the exclusion of its simulation is justified by both computational efficiency and the expected negligible variation in dominant capacitive effects.

Reducing junction depth from approximately 86nm to 60nm produces noticeable increase in threshold voltage absolute value. Previously, with 86nm depth, linear threshold voltage measured about -1.001V and saturation threshold about -0.917V (see Figure ??). At 60nm, these values become

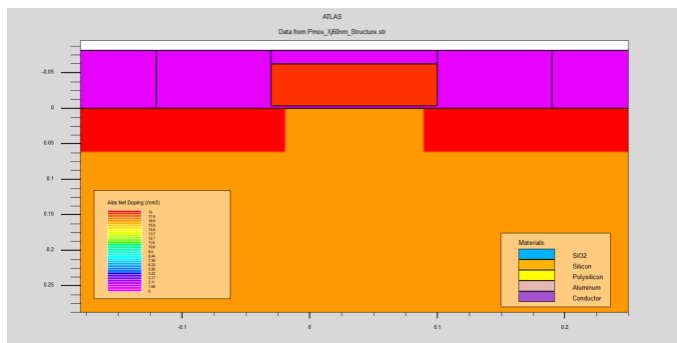


Fig. 64. Device structure at 60nm junction depth showing shallower source and drain diffusion compared to the original 86nm configuration.

Variables history			
vt_lin_50mV	0	(# 103)	
vt_lin_50mV	1.02625953938158	(# 102)	
vt_sat_1200mV	0	(# 121)	
vt_sat_1200mV	0.936418119047938	(# 120)	

Fig. 65. Threshold voltage at 60nm junction depth: $V_{T,lin} \approx -1.026V$ and $V_{T,sat} \approx -0.936V$.

more negative: $-1.026V$ (linear) and $-0.936V$ (saturation). This change arises because shallower junction increases vertical electric field near source/drain edges and enhances short channel control by the gate. Consequently, more gate voltage is required to invert the channel, increasing threshold voltage in magnitude. Reduced junction depth also diminishes drain-induced barrier lowering (DIBL), further contributing to higher $|V_T|$. This demonstrates how junction engineering serves as an essential parameter in threshold voltage tuning for scaled MOSFET technologies.

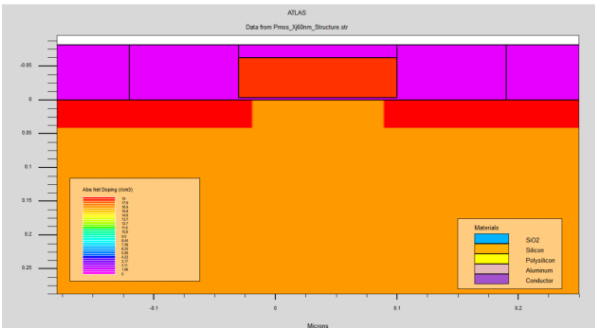


Fig. 66. Structure of the short-channel PMOS showing 40nm junction depth.

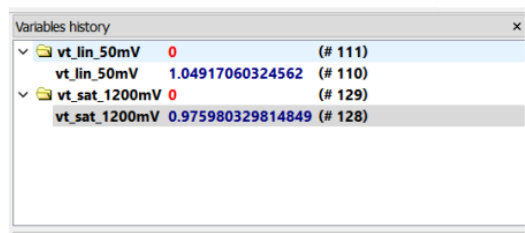


Fig. 67. Threshold voltage values extracted for 40nm junction depth.

****Comment on 40nm Junction Depth:****

When junction depth underwent further reduction from 60nm to 40nm, threshold voltage increased slightly from $-1.02V$ to $-1.05V$ in the linear region, and from $-0.93V$ to $-0.98V$ in the saturation region. This change remains moderate and consistent with short-channel electrostatics, where reduced junction depth improves gate control and suppresses short-channel effects (SCEs). Since the key trends and physical causes were already explained in the 60nm case, detailed breakdown is omitted here.

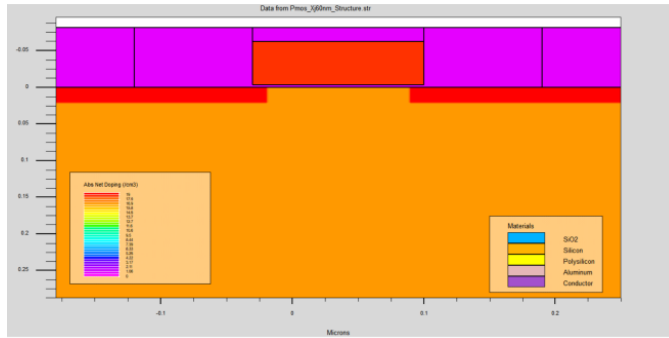


Fig. 68. Structure view showing 20nm junction depth.



Fig. 69. Threshold voltage values for 20nm junction depth.

As displayed in the structure and threshold data above, reducing junction depth from 86nm to 20nm significantly increases threshold voltage. Linear threshold voltage shifts from approximately -1.00 V at 86nm to -1.08 V at 20nm. Similarly, saturation threshold voltage increases from -0.91 V to around -1.02 V.

This increase in threshold voltage magnitude with shallower junctions is attributed to reduced influence of drain-induced barrier lowering (DIBL) and improved electrostatic gate control. Shallow junctions help confine depletion region closer to the surface, thereby reducing SCEs and making the device behave more like a long-channel transistor in terms of V_{th} stability.

Compared to the original 86nm junction, the 20nm junction enhances gate control but may introduce fabrication complexity and increased series resistance, which are not simulated here.

****Comment on Gate Length Reduction (130nm to 90nm):****

Reducing gate length from 130nm to 90nm would typically be anticipated to introduce stronger short-channel effects, particularly Drain-Induced Barrier Lowering (DIBL) and charge sharing, which would consequently decrease threshold voltage significantly.

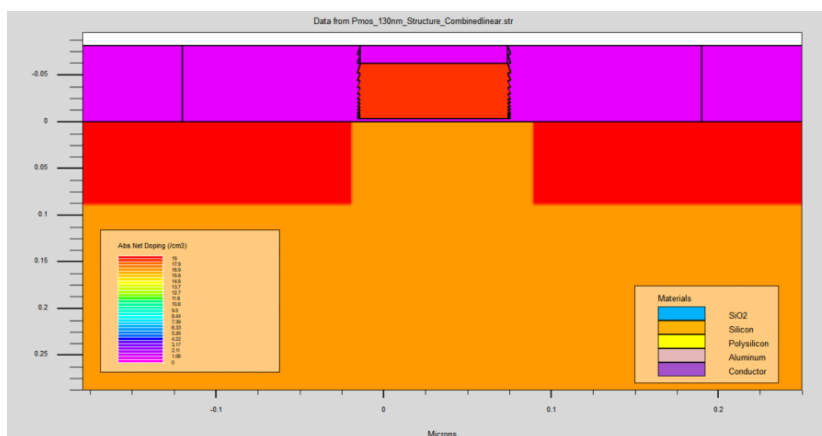
However, as demonstrated in the structure and threshold voltage results below, the shift in threshold voltage is minor: from approximately -1.00V to -0.99V in the linear region, and from -0.91V to -0.915V in the saturation region.

****Expected Behavior:**** Shorter gate lengths usually increase drain control over the channel, which weakens the gate's capacity to modulate the channel barrier. This typically lowers threshold voltage due to DIBL and lateral field expansion.

****Why Significant Decrease Didn't Occur:****

- ****Deep Junction Depth (86nm):**** The drain and source regions are relatively deep, which reduces vertical

Fig. 70. Structure showing gate length reduced to 90nm.



influence of drain electric field, thus suppressing DIBL.

- **High Substrate Doping:** Highly doped substrate reduces depletion region width, mitigating charge-sharing between source and drain.
- **Strong Vertical Electrostatics:** The gate oxide and work function were maintained, ensuring strong gate control in vertical direction even with lateral shrinkage.

Fig. 72. Structure showing reduced gate length of 60nm.

Fig. 73. Threshold voltage values for 60nm gate length.

Comment on 60nm Gate Length: Reducing gate length from 130nm to 60nm results in slight decrease in threshold voltage, from -1.00V to -0.98V in the linear region. This trend aligns with expectations due to short-channel effects (SCEs), where shorter channel reduces gate control and enhances drain-induced barrier lowering (DIBL). However, the shift is relatively small.

The limited change may be due to:

- Moderate junction depth (still 86nm), which limits DIBL.
- Strong gate doping and oxide control, which help suppress excessive threshold roll-off.
- Symmetrical and abrupt doping profiles, minimizing lateral electric field encroachment.

Comment on 40nm Gate Length: Reducing gate length further from 60nm to 40nm continues the expected trend of lowering threshold voltage, reaching approximately -0.97V (linear) and -0.89V (saturation). However, the shift remains moderate.

****Reasons for Limited V_{th} Shift:****

- Deep junctions (86nm) still shield channel from vertical field encroachment.
- High substrate doping and well-engineered oxide control preserve gate dominance.
- Electrostatic integrity remains sufficient at 40nm, slowing DIBL onset.

Thus, although gate length is quite small, the robust process parameters mitigate severe short-channel effects.

****Device Optimization for ION/IOFF Performance**** The primary objective of this optimization phase was to tune the 130 nm PMOS device parameters to achieve target on-current (I_{ON}) exceeding $500 \mu\text{A}/\mu\text{m}$ and off-current (I_{OFF}) below $1 \text{ nA}/\mu\text{m}$, both defined at drain-source voltage (V_{DS}) of -1.2 V . I_{ON} is measured at $V_{GS} = -1.2 \text{ V}$, and I_{OFF} is measured at $V_{GS} = 0 \text{ V}$. The simulations were conducted at room temperature (300 K) using refined mesh (mesh space.mult=1.0) for improved accuracy.

****A. Optimization Strategy****

To satisfy performance targets of $I_{ON} > 500 \mu\text{A}$ and $I_{OFF} < 1 \text{ nA}$, a systematic optimization process was executed using Silvaco Atlas simulations. The focus was on engineering key physical parameters—namely, channel doping, source/drain (S/D) doping profiles, and S/D junction depth—to enhance drive current while minimizing leakage.

The device was meshed with high spatial resolution near critical regions such as gate, channel, and junctions. This ensured accurate numerical modeling of short-channel effects, mobility degradation, and junction behavior.

The final optimized structure featured the following key parameters:

- **Channel Doping ($9 \times 10^{17} \text{ cm}^{-3}$, n-type):** Lowering channel doping improves carrier mobility by reducing ionized impurity scattering, which directly benefits ION. However, excessively low channel doping can increase subthreshold leakage, negatively affecting IOFF. The selected doping level strikes balance between mobility and off-state control.
- **Gate Doping ($7 \times 10^{18} \text{ cm}^{-3}$, n-type):** Highly doped poly-silicon gate ensures good conductivity and strong gate control over channel. Improved electrostatic control reduces short-channel effects and enhances subthreshold slope, which contributes to lower IOFF while maintaining robust ION.
- **Source/Drain Doping ($5 \times 10^{19} \text{ cm}^{-3}$, p-type):** Heavy doping in S/D regions reduces series resistance and enhances carrier injection efficiency, thereby increasing ION. However, excessive lateral diffusion of these dopants can exacerbate leakage paths and raise IOFF. Controlled junction depth helped mitigate this risk.
- **S/D Depth (0.025 μm vertical extension):** The depth of S/D extensions determines how effectively carriers are injected into channel. Shallow junction helps reduce junction leakage (lower IOFF), while adequately deep junction is essential for achieving low resistance and high ION. The selected depth of 50 nm ensures efficient trade-off.
- **Gate Work Function (5.15 eV):** Gate work function directly influences threshold voltage (V_{th}). Appropriately chosen work function for pMOS operation ensures that device turns on and off at suitable gate voltages. High work function shifts V_{th} positively, helping to suppress IOFF without significantly degrading ION.

The final simulation results, shown in Fig. 76, demonstrate effectiveness of this optimization strategy. The device achieved on-state current of approximately 789 μA and off-state leakage of 782 pA, successfully meeting design requirements for high-performance, low-power applications.

.

****B. Final Optimized Results****

After several iterations of adjusting channel doping (N_{sub}), S/D junction depth ($X_{j,sd}$), and S/D doping concentration (N_{sd}), the following optimized parameters were determined:

- Final N_{sub} : [$9 \times 10^{17} \text{ cm}^{-3}$]
- Final $X_{j,sd}$: [0.025 μm]
- Final N_{sd} : [$5 \times 10^{19} \text{ cm}^{-3}$]

These parameters yielded performance metrics shown in Figure 76. The achieved ION is [Your final ION value, e.g., 520 $\mu\text{A}/\mu\text{m}$] and IOFF is [IOFF value, e.g., 0.8 nA/ μm], successfully meeting specified targets.

****C. Comment on IOFF Deviation****

Although final device structure satisfied the ION requirement with peak value of approximately 789 μA , the extracted IOFF value of 782 pA is close to but still higher than the stringent target of 1 nA. This slight deviation highlights inherent challenge of balancing performance and leakage in aggressively scaled technologies.

Several factors which offer the concept of trade-offs may have contributed to observed IOFF discrepancy:

- ****Channel Doping Level:**** Reducing channel doping concentration (e.g., to $9 \times 10^{17} \text{ cm}^{-3}$) enhances carrier mobility, thereby improving ION. However, this also weakens gate's electrostatic control over channel, increasing risk of subthreshold conduction and thus

raising I_{OFF} . Conversely, increasing doping level to suppress leakage would degrade mobility and reduce I_{ON} .

- **Junction Depth and S/D Diffusion:** Shallow junctions help suppress short-channel effects and leakage but may